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Zinc is a Key Regulator of the Sperm-Specific K⁺ Channel (Slo3) Function

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This study presents a **valuable** finding on the role of intracellular zinc as a regulator of the sperm-specific potassium channel Slo3, demonstrating that zinc export during capacitation contributes to alkalization-induced membrane hyperpolarization. The electrophysiological evidence supporting zinc-mediated inhibition of Slo3 is **solid**, though the mechanistic basis of this inhibition is not complete, as the proposed zinc-binding site involving E169 and E205 has not been directly tested through double-mutant analysis. This work will be of interest to reproductive biologists and ion channel biophysicists studying the molecular mechanisms of sperm capacitation.

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Abstract

The voltage- and pH-gated Slo3 potassium channel is exclusively expressed in mammalian spermatozoa. Its sensitivity to both voltage and alkalization plays a crucial role in sperm fertility, which is tightly coupled to the capacitation process. Here we show that sperm-enriched divalent cation Zn²⁺ undergoes dynamic alteration in spermatozoa during capacitation. We also found that intracellular Zn²⁺ regulates alkalization-induced hyperpolarization in mouse spermatozoa which is mediated by Slo3 channel. Further examination of zinc regulation in mouse Slo3 (mSlo3) revealed that, in *Xenopus* oocyte expression system, intracellular zinc directly inhibits mouse Slo3 currents in dose-dependent manner at micromolar concentrations, with exceptionally slow dissociation. By combining MD simulations and electrophysiology, we also identified amino acid residues contributing to the Zn²⁺ slow dissociation from Slo3 channels. Our studies uncover the importance of intracellular zinc dynamics and its regulatory role in ion channels during sperm capacitation.

Introduction

Zinc is an essential metal ion involved in numerous biological functions, including as a regulatory component to various proteins such as enzymes, transcription factors, and ion channels (Fayyazuddin et al., 2000 [DOI](#); Gao et al., 2017 [DOI](#); Maret, 2013 [DOI](#); Sun et al., 2007 [DOI](#); Vallee & Falchuk, 1993 [DOI](#); Zheng et al., 2001 [DOI](#)). In the reproductive system, spermatozoa is a type of cell that is known to have high intracellular zinc content, which influences its production, maturation, and physiological functions (Chia et al., 2000 [DOI](#); Fallah et al., 2018 [DOI](#); Henkel et al., 1999 [DOI](#); Riffo et al., 1992 [DOI](#); Yamaguchi et al., 2009 [DOI](#)). Zinc is also present in the seminal fluid at high concentrations (~1-1.5 mM) (Henkel et al., 1999 [DOI](#); Riffo et al., 1992 [DOI](#)), and extensive studies have elucidated their role in regulating sperm physiology. For instance, extracellular zinc has been shown to influence

human sperm motility, which negatively correlates with the concentration (Henkel et al., 1999; 2005). Another study also suggested that increased zinc concentration inhibits sperm motility (Riffo et al., 1992). Despite considerable evidence that sperm motility is regulated by extracellular zinc, intracellular zinc has also been implicated in playing a regulatory role. One study demonstrated that chelating intracellular zinc reduces sperm motility and velocity (Sørensen et al., 1999). However, the underlying mechanism by which zinc regulates sperm motility remains unclear.

During fertilization, spermatozoa undergo capacitation, a series of physiological processes that determine the success of fertilizing the egg within the female reproductive tract (Austin, 1951; Chang, 1951). This capacitation involves the changes of activities in various ion channels, which leads to diverse physiological changes, such as pH, calcium level, and membrane potential (Arnoult et al., 1999; Breitbart, 2002; Publicover et al., 2007; Vredenburg-Wilberg & Parrish, 1995; Y. Zeng et al., 1995, 1996). Moreover, recent studies have shown that the activity of ion channels is regulated by zinc. Indeed, extracellular zinc has been shown to inhibit the flagellar voltage-gated proton channel Hv1, which is supposed to control intracellular pH in human sperm (Grahn et al., 2023; Lishko et al., 2010; Zhao et al., 2018). Thus, it is possible that the high levels of zinc in spermatozoa regulates their physiological properties.

In the present study, we find that intracellular zinc is exported during capacitation, indicating that zinc dynamics in spermatozoa play an important role in fertilization. Furthermore, zinc significantly inhibited the alkalization-induced hyperpolarization in mouse sperm, which is mediated by Slo3 (K_{Ca}5.1), a sperm specific pH- and voltage-gated potassium channel (Schreiber et al., 1998; X. Zhang et al., 2006).

We further discovered that intracellular zinc directly inhibits mouse Slo3 (mSlo3) currents using *Xenopus* oocytes. Of note, this inhibition is pH-dependent, voltage-dependent and has a long-lasting inhibitory effect even after the washout. Coexpression of mSlo3 with the auxiliary subunit Lrrc52, to mimic the native environment (Yang et al., 2011; X. H. Zeng et al., 2015), consistently demonstrates zinc's inhibitory effect. Additionally, we identify potential key amino acids in the mSlo3 channel that serve as the zinc binding site. Our findings highlight the importance of intracellular zinc dynamics and its regulatory role over ion channels during capacitation in spermatozoa.

Results

Intracellular zinc is exported from sperm during capacitation

To investigate the role of intracellular zinc during capacitation, we monitored the intracellular zinc concentration utilizing the zinc-sensitive fluorophore FluoZin3-AM. Fluorescence signals were successfully detected in both the head and flagellum of isolated mouse spermatozoa (Fig. 1A). As expected, the head of the spermatozoa showed a stronger fluorescence signal compared to the flagellum, likely due to its higher zinc capacity. We also treated spermatozoa with Toyoda, Yokoyama, Hoshi (TYH) medium (Toyoda et al., 1971), which is well established capacitation-inducing medium *in vitro*. We observed that there was a gradual and significant decrease in fluorescence intensity in both regions (Fig. 1B), particularly prominent in the flagellum (Fig. 1C). This decline suggests the active release of intracellular zinc from sperm flagellum occurs during capacitation. In contrast, the fluorescence intensity of the control group of non-capacitated sperm remained unchanged (Fig. 1B).

Additionally, we examined the effect of intracellular zinc on sperm motility before and after capacitation (Fig. 1—figure supplement 1A-E). We incubated the isolated spermatozoa with cell permeable Zn²⁺ chelator N,N,N',N'-Tetrakis(2-pyridylmethyl)ethylenediamine (TPEN) and measured the motility parameters before and after capacitation. We found that VAP (average path velocity), VCL (curvilinear velocity), and VSL (straight-line velocity) were influenced by the TPEN treatment only after the capacitation, as shown in Fig. 1—figure supplement 1. These results demonstrate that the dynamics of zinc levels during capacitation potentially contributes to sperm motility, highlighting the importance of zinc action in sperm physiology.

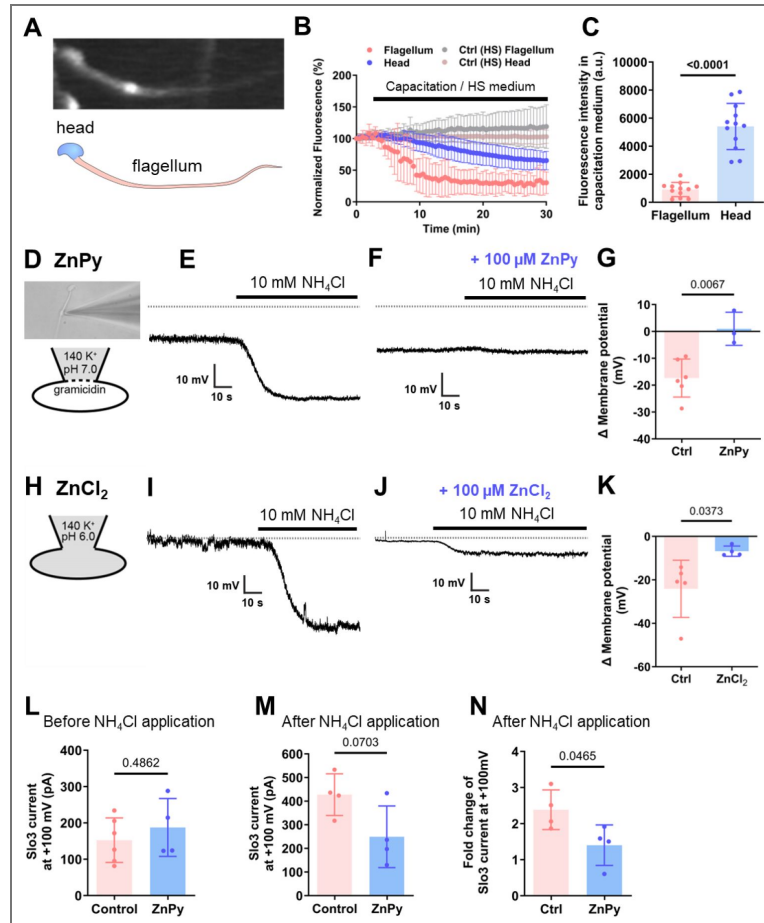


Figure 1. Intracellular zinc is exported from sperm during capacitation and inhibits the alkalinization-induced hyperpolarization in sperm.

(A-C) Intracellular zinc is exported from sperm during capacitation. **(A)** Representative image of FluoZin3-AM fluorescence observed in spermatozoa (head and flagellum, as indicated). **(B)** Time course of normalized FluoZin3-AM fluorescence response upon the application of capacitation medium. Both head and flagellum showed a decreased fluorescence response, indicating that intracellular zinc was exported from sperm in response to the application of capacitation medium. **(C)** Comparison of the fluorescence intensity from flagellum (908.6 ± 508.2 ; $n=12$) and head (5409 ± 1648 ; $n=12$) after capacitation. Unpaired t-test, $p < 0.0001$. All error bars are $\pm$ SD centered on the mean. **(D-H)** Intracellular zinc inhibits the alkalinization-induced hyperpolarization in sperm. **(D-G)** Membrane potential (V_m) recording by perforated patch-clamp were performed in mouse spermatozoa as depicted in **(D)**. **(E)** Representative trace of V_m recording for control group. The application of NH_4Cl increased the intracellular pH (pHi) which then hyperpolarized the membrane, the process called alkalinization-induced hyperpolarization. Application of NH_4Cl alone hyperpolarized the membrane by -17.37 ± 7.09 mV ($n=6$). **(F)** Representative trace of V_m recording in the presence of zinc ionophore (zinc pyrithione, ZnPy). When $100 \mu\text{M}$ ZnPy was applied on the top of NH_4Cl application, no hyperpolarization was observed ($\Delta V_m = 0.99 \pm 6.17$ mV; $n=3$). All the dotted lines indicated 0 mV level. **(G)** Comparison of ΔV_m recording between control ($n=6$) and in the presence of $100 \mu\text{M}$ ZnPy ($n=3$), $p=0.0067$, unpaired t-test. **(H-K)** V_m recording by whole-cell patch-clamp configuration in mouse spermatozoa as depicted in **(H)**. **(I)** Representative trace of V_m recording for control group. The application of NH_4Cl increased the $[\text{pHi}]$ which then hyperpolarized the membrane ($\Delta V_m = -24.12 \pm 13.16$ mV; $n=5$). **(J)** Representative trace of V_m recording in the presence of $100 \mu\text{M}$ ZnCl_2 . When zinc was applied to the intracellular part of spermatozoa on top of NH_4Cl application, alkalinization-induced hyperpolarization was inhibited ($\Delta V_m = -6.80 \pm 2.35$ mV; $n=4$). All the dotted line indicated 0 mV level. **(K)** Comparison of ΔV_m recording between control ($n=5$) and in the presence of $100 \mu\text{M}$ ZnCl_2 ($n=4$), $p=0.0373$, unpaired t-test. **(L-N)** Slo3 current recording using perforated patch-clamp in sperm. **(L)** Comparison of Slo3 current amplitude at $+100$ mV before NH_4Cl application between control (152.4 ± 25.05 pA; $n=6$) and in the presence of $100 \mu\text{M}$ ZnPy (187.5 ± 79.53 pA; $n=4$), $p=0.4862$, Welch's t-test. **(M)** Comparison of Slo3 current amplitude at $+100$ mV after NH_4Cl application between control (427.6 ± 88.28 pA; $n=4$) and in the presence of $100 \mu\text{M}$ ZnPy (249.2 ± 130.6 pA; $n=4$), $p=0.0703$, Welch's t-test. **(N)** Comparison of the fold change in Slo3 current amplitude at $+100$ mV after NH_4Cl application between control (2.39 ± 0.55 ; $n=4$) and in the presence of $100 \mu\text{M}$ ZnPy (1.4 ± 0.56 ; $n=4$), $p=0.0465$, Welch's t-test. All error bars are $\pm$ SD centered on the mean.

Intracellular zinc inhibits the alkalinization-induced hyperpolarization in sperm

It is already known that sperm capacitation is well associated with the increase of intracellular pH (Vredenburg-Wilberg & Parrish, 1995; Y. Zeng et al., 1996), which leads to the hyperpolarization of the membrane (Arnoult et al., 1999; Y. Zeng et al., 1995) as well as the elevation of intracellular Ca^{2+} concentration level (Breitbart, 2002; Publicover et al., 2007) through diverse ion channel activities. To explore whether these pathways are influenced by intracellular zinc, we used perforated patch-clamp techniques. It has been reported that under the whole-cell current clamp of mouse epididymal spermatozoa, resting membrane potential (V_m) is hyperpolarized after intracellular alkalinization (Navarro et al., 2007). Therefore, we applied 10 mM NH_4Cl , which is routinely used to alkalinize intracellular pH (Madshus, 1988), to spermatozoa. As expected, we observed that it hyperpolarized the membrane by -17.37 ± 7.09 mV; $n=6$ (Fig. 1E, G). Then, we applied 10 mM NH_4Cl with 100 μM zinc pyrithione (ZnPy), a zinc ionophore. In contrast to the control group, there was no remarkable hyperpolarization in the presence of ZnPy as the ΔV_m was 0.99 ± 6.17 mV; $n=3$ (Fig. 1F, G). These findings indicate that either extracellular or intracellular zinc inhibits the alkalinization-induced hyperpolarization in mouse sperm.

To examine the possibility that intracellular, but not extracellular, zinc is responsible for inhibiting the alkalinization-induced hyperpolarization, we measured membrane potential of sperm using the patch-clamp technique in conventional whole-cell configuration with $\text{pH}_i = 6.0$ (Fig. 1H). In the test group, the patch pipette contained 100 μM ZnCl_2 (Fig. 1J), while the control group did not (Fig. 1I). Consistent with the perforated patch-clamp results, we observed alkalinization-induced hyperpolarization upon applying NH_4Cl in the control ($\Delta V_m = -24.12 \pm 13.16$ mV; $n=5$; Fig. 1I, K). In contrast, alkalinization-induced hyperpolarization was reduced in the 100 μM ZnCl_2 group ($\Delta V_m = -6.80 \pm 2.35$ mV; $n=4$; Fig. 1J, K).

We also measured Slo3 current using perforated patch-clamp recordings in spermatozoa treated with ZnPy, before and after the addition of NH_4Cl . Control experiments were conducted in the absence of ZnPy, in which Slo3 current were recorded before and after the application of 10 mM NH_4Cl (Fig. 1L-N; Fig. 1—figure supplement 2A, B). Slo3 current in sperm tended to be inhibited by zinc, as shown by the plot of absolute Slo3 current after the addition of 10 mM NH_4Cl in the absence of ZnPy (control) and in the presence of 100 μM ZnPy (Fig. 1L, M). There was a decrease in the fold change calculated from the absolute current before and after the addition of 10 mM NH_4Cl of ZnPy treated group (1.4 ± 0.56 ; $n=4$) compared to the control group (2.39 ± 0.55 ; $n=4$; Fig. 1N). Taken together, these results confirmed that intracellular zinc indeed inhibits alkalinization-induced hyperpolarization in mouse sperm.

Zinc inhibits mouse Slo3 (mSlo3) current

Our findings suggest that intracellular zinc inhibits a key process in sperm capacitation, specifically the alkalinization-induced hyperpolarization. Previous studies have identified the pH- and voltage-dependent potassium channel Slo3 is responsible for the principal K^+ current ($I_{\text{K}_{\text{Sper}}}$) in mouse spermatozoa (Navarro et al., 2007; Santi et al., 2010; Schreiber et al., 1998; X. H. Zeng et al., 2011). During capacitation, the rise in pH_i leads to the activation of Slo3 channels, resulting in membrane hyperpolarization (Santi et al., 2010). Given this context, we next investigated whether intracellular zinc acts directly on the Slo3 channel.

We expressed the mSlo3 channel in the *Xenopus* oocyte expression system and conducted inside-out patch-clamp recordings. The bath solution was initially set to pH 8.0, and recordings were performed using a high-speed perfusion system. A ramp pulse from -100 mV to +100 mV with a holding potential of -60 mV was applied. We observed a robust pH- and voltage-dependent mSlo3 current at pH 8.0 (Fig. 2A, B), consistent with previous reports (Leonetti et al., 2012; Schreiber et al., 1998; X. Zhang et al., 2006). The application of 100 μM of ZnCl_2 resulted in the inhibition of $36.62 \pm 11.65\%$ ($n=6$) of mSlo3 current (Fig. 2A, B). Of note, the inhibited mSlo3 current by zinc was never restored after the wash out of zinc even for more than 30 seconds. However, when

pH 6.0 was applied to completely abolish the mSlo3 current, followed by the application of pH 8.0, the mSlo3 current recovered (Fig. 2A, B). We also determined the IC_{50} of zinc inhibition by applying zinc at concentrations 0.1, 1, 10, 100, and 1000 μ M for 30 seconds. The dose-response curve, fitted to the Hill equation with a standard slope (Hill slope of -1.0), yielded an IC_{50} = 145.2 μ M (Fig. 2C).

We also tested the effect of zinc at a lower pH of 7.5. We observed the pH- and voltage-dependent mSlo3 current at this pH. However, the 100 μ M of $ZnCl_2$ application did not affect the mSlo3 current (Fig. 2D-F). To rule out instrumental and human error, we also tested the effect of pH 6.0 which suppresses Slo3 activity. As expected, the application of pH 6.0 completely inhibited the mSlo3 current, and the subsequent application of pH 7.5 restored the current (Fig. 2D-F).

We also conducted step pulse protocols from -100 mV to +180 mV with a holding potential of -60 mV to assess whether zinc inhibition is voltage-dependent. At pH 8.0, we observed a robust mSlo3 current, which was inhibited by 100 μ M $ZnCl_2$ (Fig. 2G, H). The current-voltage (I-V) relationship indicated voltage-dependent zinc inhibition (Fig. 2H). The ratio of I_{zinc} to $I_{control}$ plotted against membrane potential also demonstrated that zinc inhibition of mSlo3 current is voltage-dependent (Fig. 2—figure supplement 1B).

In addition, we plotted the conductance-voltage (G-V) relationship by calculating the conductance from the maximum current amplitude at steady-state, as the tail current recordings of mSlo3 were unsuccessful. We noted that V_{50} of mSlo3 current shifted to more hyperpolarized potentials in the presence of zinc (V_{50} control = 80.67 ± 2.32 ; V_{50} zinc = 49.68 ± 1.38 ; $n=13$; Fig. 2—figure supplement 1C).

As we observed the long-lasting inhibitory effect of zinc on mSlo3 current, we examined whether a zinc chelator could restore the mSlo3 current to the baseline level without applying acidic pH. We used 5 mM EGTA, a divalent cation chelator which strongly chelates zinc ($Kd_{(Zn)} \sim 1$ nM) (Hu et al., 2009). Zinc-inhibited mSlo3 current could not be recovered by pH 8.0 alone (Fig. 2I). However, significant current recovery was observed when 5 mM EGTA was included in the pH 8.0 bath solution (Fig. 2J). Specifically, 5 mM EGTA recovered 49.22 ± 37.98 % of the mSlo3 current ($n=5$), compared to pH 8.0 alone, which showed no significant recovery (5.62 ± 24.4 %; $n=7$), as shown in Fig. 2K.

Additionally, we investigated the presence of endogenous zinc in Slo3 by perfusing pH 8.0 followed by 5 mM EGTA at the beginning of the recording to detect any changes in the baseline current. The baseline current showed a slight increase upon perfusion with 5 mM EGTA, indicating the presence of endogenous zinc affecting the baseline current in Slo3 (Fig. 2—figure supplement 1D, E).

Taken together, these results showed that zinc has a long-lasting inhibitory effect on mSlo3 current that cannot be reversed by pH 8.0 alone. The inhibitory effect is pH-dependent, as complete inhibition at pH 6.0 followed by recovery at pH 8.0 was observed. In addition, zinc inhibition is voltage-dependent, and the application of 5 mM EGTA, which strongly chelates zinc, can partially recover mSlo3 current.

Zinc more efficiently inhibits mSlo3 current when coexpressed with the auxiliary subunit Lrrc52

In its native environment of spermatozoa, Slo3 is expressed with its auxiliary subunit Lrrc52 (Yang et al., 2011; X. H. Zeng et al., 2015). Lrrc52 shifts the Slo3 gating to more negative potentials, making the channel easier to open (Yang et al., 2011). We coexpressed mSlo3 with mouse Lrrc52 (mLrrc52) in the *Xenopus* expression system to perform inside-out patch clamp recordings.

The mSlo3+mLrrc52 current was observed upon the application of pH 8.0. A ramp pulse protocol from -100 mV to +100 mV was applied with a holding potential of 0 mV. Similar to mSlo3 alone, in the presence of mLrrc52, mSlo3 current was also inhibited by 100 μ M $ZnCl_2$ (Fig. 3A, B). In addition, zinc inhibition was not recovered after washing with the zinc-free solution (Fig. 3B). A

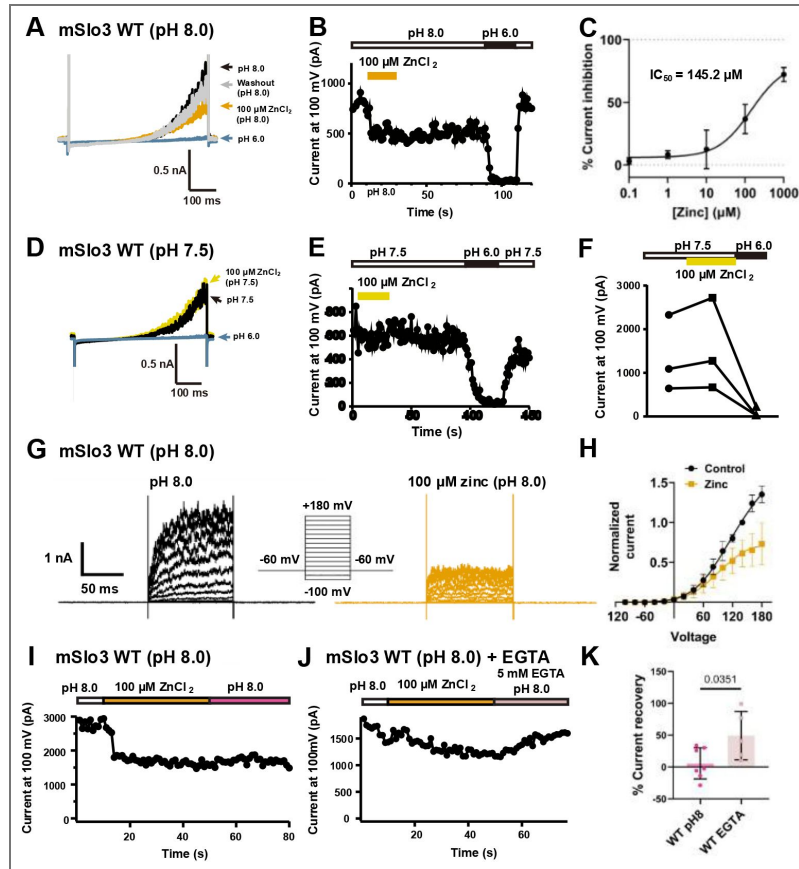


Figure 2. Zinc inhibits mouse Slo3 (mSlo3) current.

(A-C) Inside-out patch-clamp recording of mSlo3 with $\text{pHi}=8.0$ by applying ramp pulse from -100 mV to $+100$ mV with the holding potential of 0 mV. (A) Representative current traces upon the application of pH 8.0 (black); $100 \mu\text{M}$ of zinc in pH 8.0 (yellow orange); pH 6.0 (blue); and wash-out by pH 8.0 (gray) after pH 6.0. (B) Time course of the change in current amplitude at $+100$ mV in response to pH 8.0, $100 \mu\text{M}$ zinc, pH 6.0, and pH 8.0 as indicated (white: pH 8.0; yellow orange: $100 \mu\text{M}$ zinc in pH 8.0; solid black: pH 6.0). (C) Dose-response curve of zinc inhibition in mSlo3 current (pH 8.0). % current inhibition was plotted assuming that the dose response curve has a standard slope, equal to a Hill slope of -1.0 . $\text{IC}_{50} = 145.2 \mu\text{M}$; $n = 2-6$. (D-F) Inside-out patch-clamp recording of mSlo3 in *Xenopus* oocyte with $\text{pHi}=7.5$ by applying ramp pulse from -100 mV to $+100$ mV with the holding potential of 0 mV. (D) Representative current traces upon the application of pH 7.5 (black); μM of zinc (yellow); and pH 6.0 (blue). (E) Time course of the change in current amplitude at $+100$ mV in response to pH 7.5, $100 \mu\text{M}$ zinc, and pH 6.0 as indicated (white: pH 7.5; yellow: $100 \mu\text{M}$ zinc in pH 7.5; solid black: pH 6.0). (F) Current amplitude at $+100$ mV upon the application of pH 7.5, $100 \mu\text{M}$ zinc in pH 7.5, and pH 6.0 from individual cell, as indicated. (G-H) Inside-out patch-clamp recording of mSlo3 with $\text{pHi}=8.0$ by applying step pulses from -100 mV to $+180$ mV with the holding potential of -60 mV. Zinc inhibition is voltage-dependent. (G) Representative current traces upon the application of pH 8.0 (black) and $100 \mu\text{M}$ of zinc in pH 8.0 (yellow orange). (H) I-V relationship of control (pH 8.0; black) and zinc ($100 \mu\text{M}$; yellow orange). Normalization was performed by dividing the absolute current of mSlo3 at pH 8.0 of each voltage by the absolute current at the pre-determined highest voltage that still produced a stable mSlo3 current (i.e., good patch, good clamp). In this analysis, $+140$ mV was chosen as the highest voltage for normalization, since in few cells the patch was lost at $+160$ mV and $+180$ mV. Similar to the control condition, the absolute current of mSlo3 in the presence of $100 \mu\text{M}$ zinc was normalized to the absolute current of the control at $+140$ mV. I-V curve was fitted by Boltzmann equation for control and followed by Woodhull equation for zinc inhibition ($V_{50 \text{ Ctrl}} = 85.24$; $\text{Slope}_{\text{Ctrl}} = 35.51$; $V_{\text{block Zn}} = 188.8$; $\text{Slope}_{\text{block Zn}} = -115.3$; $n=13$). (I-K) Inside-out patch-clamp recording of mSlo3 with the application of 5 mM EGTA in pH 8.0 by applying ramp pulse from -100 mV to $+100$ mV with the holding potential of 0 mV. Zinc has a long-lasting inhibitory effect on mSlo3, and it could be rescued by EGTA (I) Time course of the change in current amplitude at $+100$ mV in response to pH 8.0, $100 \mu\text{M}$ zinc, and pH 8.0 as indicated (white: pH 8.0; yellow orange: $100 \mu\text{M}$ zinc in pH 8.0; pink: pH 8.0 for wash-out). (J) Time course of the change in current amplitude at $+100$ mV in response to pH 8.0, $100 \mu\text{M}$ zinc, and 5 mM EGTA in pH 8.0 as indicated (white: pH 8.0; yellow orange: $100 \mu\text{M}$ zinc in pH 8.0; light lilac: 5 mM EGTA in pH 8.0 for wash-out). (K) Comparison of percentage of current recovery between pH 8.0 only ($5.62 \pm 24.40\%$; $n = 7$) and 5 mM EGTA in pH 8.0 ($49.22 \pm 37.98\%$; $n=5$), $p=0.0351$, unpaired t-test. All error bars are $\pm$ SD centered on the mean.

significant leftward shift in IC_{50} was observed when the dose-response curve was plotted for coexpression with mLrrc52 (Fig. 3C; $IC_{50} = 15.2 \mu M$; $n = 3-8$) compared to mSlo3 alone (Fig. 2C, Fig. 3C).

Additionally, when pH 7.5 was applied to depolarized potentials, mSlo3+mLrrc52 current was activated and in contrast with mSlo3 alone (Fig. 2D-F), 100 μM $ZnCl_2$ already inhibited mSlo3+mLrrc52 current at pH 7.5 (Fig. 3D, E, Fig. 3—figure supplement 1D). Alkalinization appeared to increase the percentage of current inhibition by 100 μM $ZnCl_2$ (Fig. 3F; pH 7.5 = $32.51 \pm 8.62\%$; $n=3$ and pH 8.0 = $52.61 \pm 8.23\%$; $n=8$).

We next conducted recordings using a step pulse protocol from -100 mV to +180 mV with a holding potential of -60 mV. Upon applying pH 8.0, we observed a significant mSlo3+mLrrc52 current, which was inhibited by 100 μM $ZnCl_2$ (Fig. 3—figure supplement 1A, B). Again, the I-V relationship confirmed that zinc inhibition was voltage-dependent (Fig. 3—figure supplement 1B). V_{50} of mSlo3+mLrrc52 current was also shifted into more hyperpolarized potentials in the presence of zinc (V_{50} control = 72.93 ± 2.06 V_{50} zinc = 56.27 ± 2.75 ; $n=13$; Fig. 3—figure supplement 1C).

Taken together, when co-expressed with mLrrc52, mSlo3 current was more efficiently inhibited by zinc.

E169 and E205 located within VSD are important for the sustained zinc inhibition of mSlo3 current

To elucidate the mechanism of zinc inhibition on the mSlo3 channel, we aimed to identify the structural determinants required for this inhibition using a range of computational tools and mutagenesis. First, we used the AlphaFold3 structure prediction webserver (Abramson et al., 2024) to predict the zinc binding site on the mSlo3 channel. Notably, the prediction has shown that the majority of the zinc binding sites are located in the channel's intracellular regulator of K^+ conductance (RCK) domain, which consists of RCK1 and RCK2, with very few zinc sites in the transmembrane domain (Fig. 4—figure supplement 1A). The RCK domain is also recognized as crucial for zinc binding in Slo channel families (Hou et al., 2010; J. Zhang et al., 2023).

To validate the site within the intracellular region, we conducted site-directed mutagenesis on intracellular domain focusing on RCK1, RCK1/RCK2 linker, and RCK2 (H354R, H383R, H423R, H516R, H564R, H606A, H695R, and H720R) to assess changes in channel sensitivity to zinc. However, the percentage of current inhibition varied across the mutated constructs, showing either increases or no appreciable change (Fig. 4—figure supplement 1B, C). We also generated a chimera where RCK2 of mSlo3 was replaced with RCK2 from mSlo1, but this chimera exhibited slightly increased sensitivity to zinc (Fig. 4—figure supplement 1B, C). Taken together, this concludes that, unlike Slo1, the zinc binding site is unlikely to be located in the RCK or the intracellular domain of the channel.

We then diverge our focus to the intracellular site of the channel. Given that zinc has been shown to shift the G-V relationship of the mSlo3 channel and its complex with Lrrc52 (Fig. 2—figure supplement 1C, Fig. 3—figure supplement 1C), we suspect that the binding site is more likely to be located in the transmembrane (TM) region. To enhance the prediction within the transmembrane segment, we computationally truncated the intracellular domain, allowing a key binding site to be found near E169 and E205 within the voltage-sensing domain (VSD) with the AlphaFold3 structure prediction webserver (Fig. 4A). Next, we conducted triplicates of 200 ns molecular dynamics simulations of the AlphaFold3 full-length Slo3 tetrameric structure to validate this binding site where the Slo3 channel is solvated in 0.15 M $ZnCl_2$ and 0.15 M KCl (Fig. 4B). Here, we show that using MD simulations, a similar zinc site is also observed near E169 and E205, validating the prediction results by AlphaFold3 (Fig. 4C). In addition, we also observed another zinc binding near the pore at E310 and E313 (Fig. 5—figure supplement 1A).

To test the stability of zinc within E169 and E205, we removed all the $ZnCl_2$ in the simulation box, leaving just 4 zinc ions near E169 residues and conducted three separate simulations for 200 ns, leading to 12 data points. Our simulations show that zinc remains stable, with the low root mean

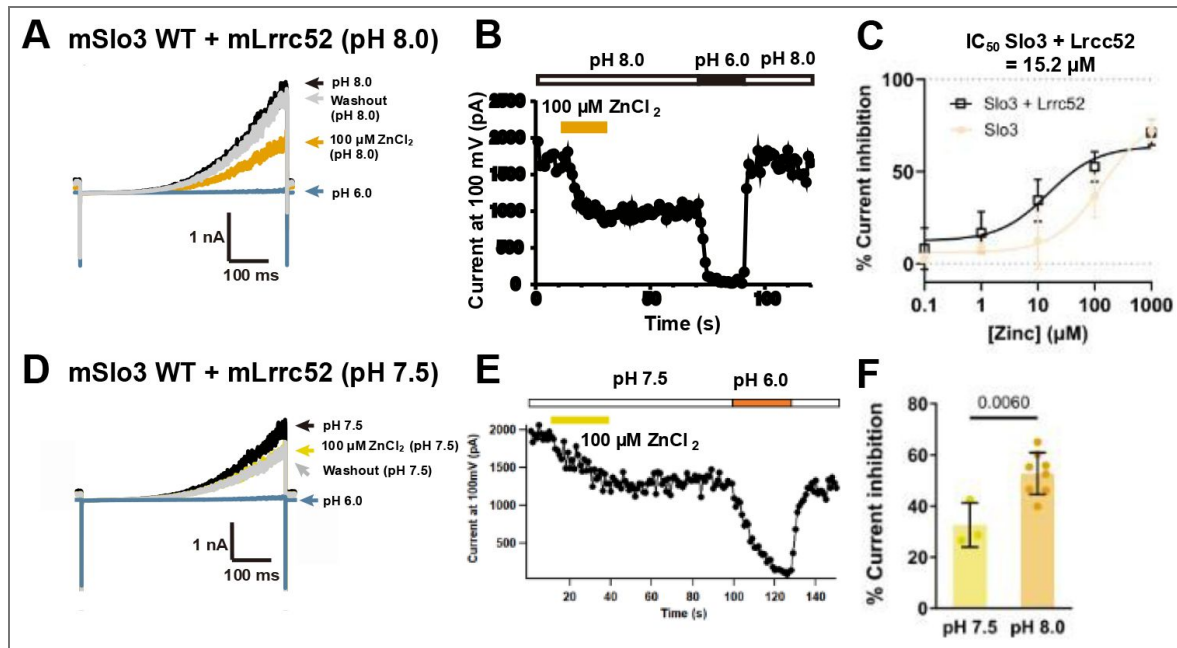


Figure 3. Zinc inhibits mSlo3 current when co-expressed with gamma subunit Lrrc52.

(A-B) Inside-out patch-clamp recording of mSlo3 co-expressed with mLrrc52 with $pH_i=8.0$ by applying ramp pulse from -100 mV to $+100$ mV with the holding potential of 0 mV. (A) Representative current traces upon the application of pH 8.0 (black); $100 \mu\text{M}$ of zinc in pH 8.0 (yellow orange); pH 6.0 (blue); and wash-out by pH 8.0 after pH 6.0 (gray). (B) Time course of the change in current amplitude at $+100$ mV in response to pH 8.0, $100 \mu\text{M}$ zinc, pH 6.0, and pH 8.0 as indicated (white: pH 8.0; yellow orange: $100 \mu\text{M}$ zinc in pH 8.0; solid black: pH 6.0). (C) Dose-response curve of zinc inhibition in mSlo3 co-expressed with mLrrc52 current (pH 8.0). % current inhibition was plotted assuming that the dose response curve has a standard slope, equal to a Hill slope of -1.0 . $IC_{50} = 15.2 \mu\text{M}$; $n = 3-8$. (D-E) Inside-out patch-clamp recording of mSlo3 in *Xenopus* oocyte co-expressed with mouse Lrrc52 (mLrrc52) with $pH_i=7.5$ by applying ramp pulse from -100 mV to $+100$ mV with the holding potential of 0 mV. (D) Representative current traces upon the application of pH 7.5 (black); $100 \mu\text{M}$ of zinc (yellow); pH 7.5 for wash-out (grey); and pH 6.0 (blue). (E) Time course of the change in current amplitude at $+100$ mV in response to pH 7.5, $100 \mu\text{M}$ zinc, pH 6.0, and pH 7.5 for wash-out as indicated (white: pH 7.5; yellow: $100 \mu\text{M}$ zinc in pH 7.5; orange pH 6.0). (F) Comparison of the percentage of current inhibition by $100 \mu\text{M}$ zinc between pH 7.5 ($32.51 \pm 8.62\%$; $n=3$) and pH 8.0 ($52.61 \pm 8.23\%$; $n=8$), $p=0.006$; unpaired t-test. All error bars are $\pm$ SD centered on the mean.

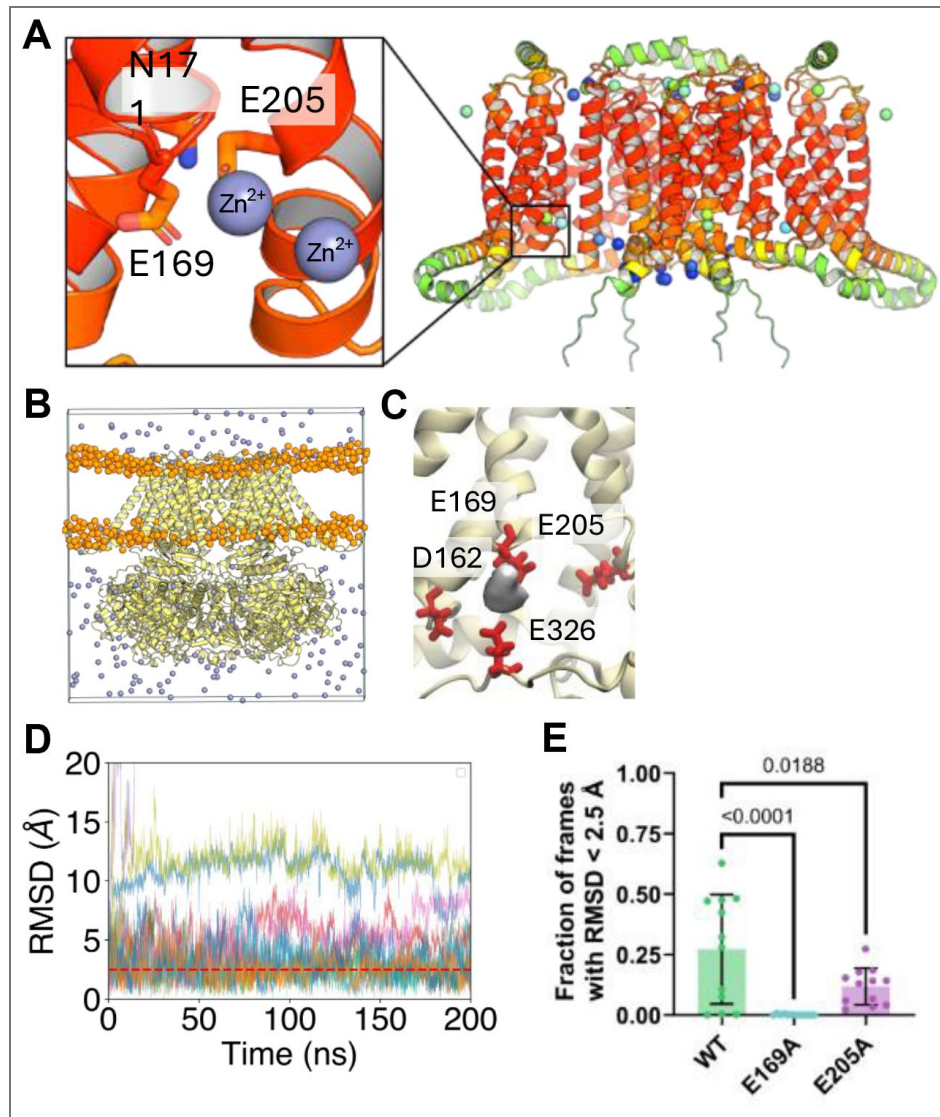


Figure 4. Computational studies show that the Zn binding site on mSlo3 is located near E169 and E205.

(A) AlphaFold3 prediction of the Zn binding site on the transmembrane segment of the Slo3 channel. Zn is shown as grey spheres, and Slo3 is shown using an orange cartoon. (B) The system is set up for flooding simulations. The phosphorus atoms on the POPC lipid are shown in orange. All Zn ions in the system are shown as grey spheres. The Slo3 channel is shown in the yellow cartoon. All water, KCl and other atoms within POPC molecules are omitted for clarity. (C) The density of Zn ions within 4 Å of the protein molecules was calculated using VolMap averaged across 200 ns flooding simulations (n = 3). Key acidic contacting residues are shown as red sticks. (D) Root mean square (RMSD) calculation of the 12 Zn ions fitted to the C alpha atoms of the protein on the first frame for 200 ns. Given that there are 4 Zn in each subunit per repeat, this leads to 12 data sets shown in different colored plots. The red dotted line marked the cutoff at 2.5 Å. (E) The fractions of frames where the RMSD of zinc ions are less than 2.5 Å in WT channel, E169A and E205A mutants (p < 0.0001, p = 0.0188, respectively). Statistical analysis was done by one-way ANOVA with Dunnett's post-hoc test compared to the WT group for each mutant (n = 12). All error bars are ± SD centered on the mean.

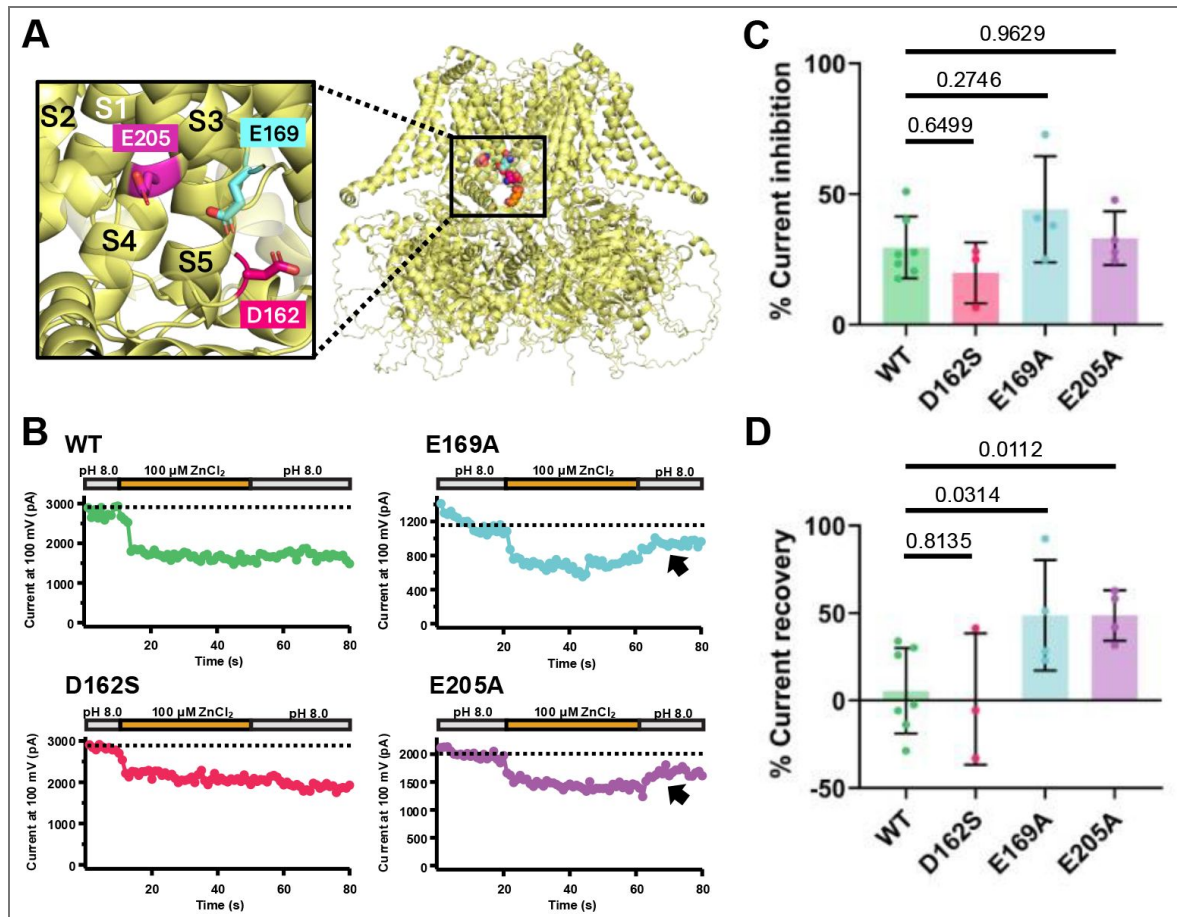


Figure 5. E169 and E205 located within VSD are important for the sustained zinc inhibition of mSlo3 current.

(A) Tetrameric mSlo3 structure representation. Inset shows rotated view of selected region which indicates predicted zinc binding site from MD simulations located within the voltage-sensor domain: D162 (S2-S3 linker), E169 (S3 domain), and E205 (S4 domain) which have been mutated to elucidate the zinc binding site in mSlo3. mSlo3 structure representation in (A) was based on structure prediction using AlphaFold3 (Abramson et al., 2024), mSlo3 sequence was retrieved from NCBI NP_032458.3. Some portions of the S2 and S3 domains have been rendered highly transparent to enhance clarity. (B) Time course of the change in current amplitude at +100 mV in response to pH 8.0, 100 μM zinc, and pH 8.0 for wash-out as indicated for: WT, D162S, E169A, and E205A. The dotted line indicated the level of activated mSlo3 current at pH 8.0. Black arrows indicated the current recovery observed upon wash-out. (C) Comparison of the percent current inhibition upon the application of 100 μM zinc between WT, D162S, E169A, and E205A ($p=0.6499$, $p=0.2746$, $p=0.9629$, respectively). Statistical analysis was done by one-way ANOVA with Dunnett's post-hoc test compared to the control group (WT), $n=3-7$. (D) Comparison of percentage of current recovery upon wash-out using pH 8.0 between WT, D162S, E169A, and E205A ($p=0.8135$, $p=0.0314$, $p=0.0112$, respectively). Statistical analysis was done by unpaired t-test compared to the WT group for each mutant, $n=3-7$. All error bars are $\pm$ SD centered on the mean.

square deviation (RMSD) within its binding site in the transmembrane region, validating the stability within its binding site (Fig. 4D [↗](#)).

To validate the binding site further, we conducted a computational site-directed mutagenesis to further assess zinc stability after the mutation of the nearby acidic residue, E169A and E205A. Here, we show that the RMSD of the E169A mutant increases more than WT and E205A (Fig. 4E [↗](#)). Together, our simulation suggested that E169 and E205 are indeed important for the binding of zinc by Slo3.

We then examined zinc sensitivities of mSlo3 currents from oocytes expressing mutants by electrophysiology with mutation of zinc binding residues on the mSlo3 channel predicted by MD simulations, which is located in the S2-S3 linker (D162), S3 domain (E169), S4 domain (E205) and the intracellular linker between TM and RCK1 (E326) (Fig. 4C-E [↗](#), Fig. 5A [↗](#), Fig. 5—figure supplement 1B [↗](#)). All of these mutants exhibited similar current inhibition by zinc compared to wildtype (WT = 29.58 ± 11.84 %, n=7; D162S = 19.85 ± 11.63 %, n=3; E169A = 44.17 ± 20.27 %, n=4; E205 = 33.08 ± 10.28 %, n=4; E326 = 27.54 ± 1.27 %, n=2; Fig. 5B, C [↗](#), Fig. 5—figure supplement 1C, D [↗](#)).

On the other hand, when we analyzed the percentage of current recovery of all the mutants, E169A and E205A showed significant current recovery upon the wash-out by pH 8.0 alone (WT = 5.62 ± 24.40 %, n=7; D162S = 0.86 ± 37.60 %, n=3; E169A = 48.74 ± 31.61 %, n=4; E205A = 48.65 ± 14.37 %, n=4; E326 = -3.27 ± 1.43 %, n=2; Fig. 5B, D [↗](#), Fig. 5—figure supplement 1C, E [↗](#)). Additionally, we performed sequence alignment by using ClustalO between mSlo3, mSlo1, and mSlo2.2. It is worth noting that only human and frog variants of Slo2.1 sequence are available in the database, so we included only Slo2.2 subtype, as our focus was on Slo3 in mouse sperm. Based on the alignment, E169 (mSlo3 numbering) is conserved among the Slo family channels in mice, while in contrast E205 (mSlo3 numbering) is not (Figure 5 – supplementary figure 1F [↗](#)).

Taken together, consistent with MD simulation results, our electrophysiological recordings demonstrated that the long-lasting inhibitory effect of zinc was partly abolished by these mutations. Thus, our findings highlight the contribution of E169A in the bottom of S3 domain and E205A in the bottom of S4 domain in zinc-mediated inhibition of mSlo3 current.

Zinc alters the dynamics of VSD motion in mSlo3

Given the observed zinc-induced shift in the G-V relationship of both mSlo3 and mSlo3+mLrrc52, along with MD simulations and mutagenesis data, it is likely that VSD is involved in zinc inhibition of mSlo3. Then, we performed voltage-clamp fluorometry (VCF) experiments to track the motion of the VSD in the presence and absence of zinc. First, to identify suitable site for labeling with TMRM, we performed VCF scanning by introducing Cys mutations one at a time at the top of S3, S3-S4 linker, and the top of S4 of mSlo3 VCF construct (Fig. 6—figure supplement 1A [↗](#)). Two native, extracellular cysteines were mutated to avoid non-specific labeling (mSlo3 C19S/C131S) and the desired Cys mutations were then added on this double-mutant background. Initial scanning of VCF constructs L188C, K189C, S190C, and N191C (Fig. 6—figure supplement 1A-E [↗](#)) conjugated with TMRM resulted in no detectable fluorescence intensity (F) changes (ΔF). Subsequently, we coexpressed the mSlo3 VCF constructs with mLrrc52 to increase the level of mSlo3 expression in heterologous expression system (Leonetti et al., 2012 [↗](#)). Out of three mutants coexpressed with mLrrc52 (L193C, G194C, and L195C; Fig. 6—figure supplement 1F-H [↗](#)), L193C, which is located in the top of S4 domain, produced fluorescence intensity changes in response to voltage steps (Fig. 6—figure supplement 1F [↗](#)) ($\Delta F/F = 1.15 \pm 0.48$ %; n=4).

We next performed VCF recordings in the absence and presence of zinc. Oocytes were held at -60 mV and the step pulses were applied from -160 mV to +160 mV. Fluorescence change could be observed in response to voltage steps ($\Delta F/F_{\text{Control}} = 2.84 \pm 1.25$ % at +160 mV; n=7; Fig. 6A [↗](#)). A puff of 10 mM zinc solution was then manually injected into the same oocyte using a glass needle with positive pressure. 60 seconds post-injection, a second set of step pulses was applied to record the response under zinc application ($\Delta F/F_{\text{Zinc}} = 2.53 \pm 1.37$ % at +160 mV; n=7; Fig. 6A [↗](#)). In the presence of zinc, the V_{50} of fluorescence-voltage (F-V) relationship was shifted into more

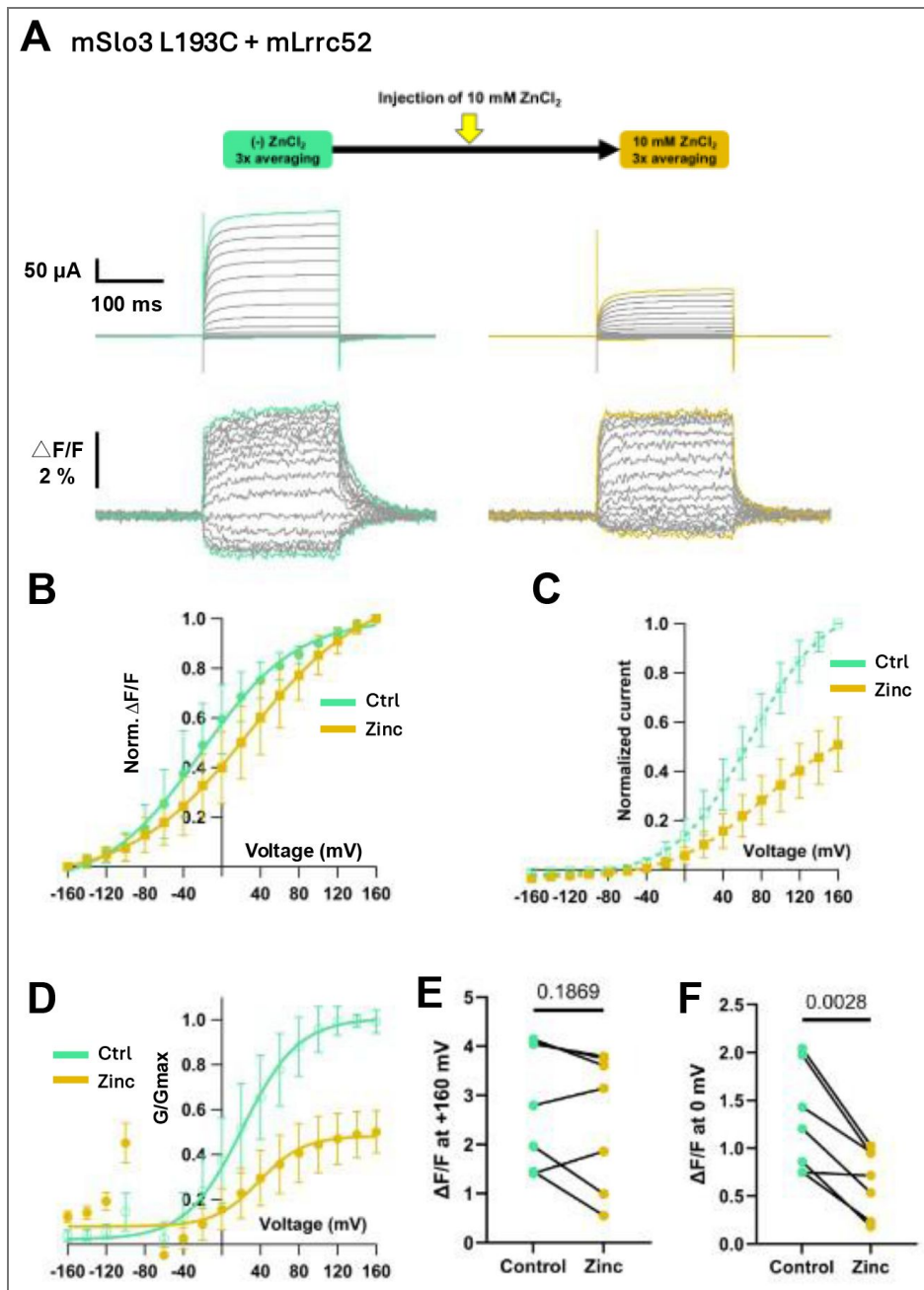


Figure 6. Zinc influenced the motion of the voltage-sensing domain (VSD) of mSlo3 channel.

(A) Representative current traces and fluorescence signals of mSlo3 L193C coexpressed with Lrrc52 in the absence (green) and presence (yellow orange) of zinc. The inset illustrates a schematic timeline of direct zinc injection using a glass needle and positive pressure. Voltage steps from -160 mV to +160 mV were applied with a holding potential of -60 mV. Zinc was injected into the same cell after the initial VCF recording (green), followed by a second VCF recording 60 seconds post-injection (yellow orange). $\Delta F/F_{\text{Control}} = 2.84 \pm 1.25\%$ ($n=7$); $\Delta F/F_{\text{Zinc}} = 2.53 \pm 1.37\%$ ($n=7$). The green and yellow orange traces represent the current and fluorescence signals at in +160 mV and -160 mV, in the absence (green) and presence (yellow orange) of zinc. (B) Comparison of the F-V relationship between control (green) and zinc (yellow orange). The F-V curve was fitted using Boltzmann equation ($V_{50} \text{ control} = -21.68 \pm 3.04$; $V_{50} \text{ zinc} = 27.04 \pm 1.05$; $n=7$). Normalization was done based on the maximum $\Delta F/F$ for each control and zinc-injected conditions. (C) I-V relationship showing inhibition of mSlo3 L193C+Lrrc52 currents by direct zinc injection. Traces compare control (green) and zinc (yellow orange). (D) G-V relationship comparison of mSlo3 L193C current co-expressed with mLrrc52 between control (green circle) and zinc-injected (yellow orange circle) conditions ($V_{50} \text{ control} = 21.11 \pm 4.8$; $V_{50} \text{ zinc} = 38.62 \pm 28.8$; $n=7$). Normalization was done based on the maximum conductance in control. (E) Comparison of $\Delta F/F$ in the absence and presence of zinc at +160 mV from the same cell ($p=0.1869$, paired t-test, $n=7$). (F) Comparison of $\Delta F/F$ in the absence and presence of zinc at 0 mV from the same cell ($p=0.0028$, paired t-test, $n=7$). All error bars are $\pm$ SD centered on the mean.

depolarized potentials (V_{50} control = -21.68 ± 3.04 ; V_{50} zinc = 27.04 ± 1.05 ; Fig. 6B). This decrease in $\Delta F/F$ was more pronounced at certain voltages, such as 0 mV (Fig. 6E-F), and extended up to +80 mV. I-V relationship (Fig. 6C) showed that mSlo3 current was inhibited by zinc, consistent with what has been observed in the patch-clamp recordings (Fig. 2 and Fig. 3). Consistent with F-V relationship, the conductance-voltage (G-V) was also shifted into more depolarized potentials (V_{50} control = 21.11 ± 4.8 ; V_{50} zinc = 38.62 ± 28.8 ; Fig. 6D). Collectively, these results clearly indicate that the VSD motion of mSlo3 is influenced by the application of intracellular zinc.

Discussion

The present study aims to understand the roles of intracellular zinc in sperm physiology, focusing on sperm capacitation. Our results demonstrated that intracellular zinc is exported from spermatozoa during capacitation. We further discovered that intracellular zinc regulates alkalization-induced hyperpolarization in mouse spermatozoa, mediated by Slo3 channel. Detailed analysis of zinc's regulatory effects mSlo3 in the *Xenopus* oocyte expression system revealed that intracellular zinc directly inhibits mSlo3 currents with exceptionally slow dissociation. By combining MD simulations and electrophysiology, we identified amino acid residues important for zinc-mediated inhibition of Slo3 activities.

Zinc levels are dynamic during capacitation

Using FluoZin3-AM for zinc imaging, we confirmed the presence of intracellular zinc in sperm (Fig. 1A), which is consistent with previous findings (Henkel et al., 1999). Our observations revealed that treatment with capacitation medium induced a decrease in zinc fluorescence intensity (Fig. 1B, C), suggesting that zinc levels are dynamic during capacitation. Previous studies using boar, bull and human spermatozoa have identified four distinct patterns of zinc localization associated with the state of capacitation (Kerns et al., 2018). In non-capacitated sperm, high concentration of zinc is present throughout the whole sperm from the sperm head to the whole sperm-tail, and the subcellular zinc localization shifts to a medium level in both the sperm head and the sperm tail mid piece (Kerns et al., 2018). In congruent with this, our findings also demonstrated a decrease in fluorescence intensity in both the head and flagellum of mouse spermatozoa during capacitation (Fig. 1B, C). These results suggest that zinc dynamics appear to play an important role in sperm capacitation.

Indeed, we observed that zinc chelator significantly affected the sperm motility specifically in VAP (average path velocity), VCL (curvilinear velocity), and VSL (straight-line velocity) only after, but not before, capacitation (Fig. 1—figure supplement 1). Of note, it has been recently reported that all these motility parameters (VAP, VCL, and VSL) are reduced by Slo3-specific inhibitors in human sperm (M. Lyon et al., 2023). These findings are consistent with the idea that endogenous zinc dynamics control sperm motility through Slo3 during the capacitation process.

The identity of the transporters directly responsible for zinc dynamics remains unclear. To maintain zinc homeostasis, two protein families facilitate the transport of Zn^{2+} across cellular and intracellular membranes in opposite directions: zinc transporter proteins (ZnT) and Zrt-, Irt-like proteins (ZIP) (Kambe et al., 2004). ZIP12 has been reported to be highly expressed in mouse testis (Zhu et al., 2022), as well as ZnT-1 (Elgazar et al., 2005). The mechanisms of zinc transport in sperm during capacitation remains a subject to be investigated in future studies.

Intracellular zinc directly inhibits mSlo3 channels

Our findings from inside-out patch-clamp recordings indicate that zinc directly inhibits mSlo3 channels in a dose-dependent manner (Fig. 2). Zinc inhibition in mSlo3 channels is dependent on pH (Fig. 2A-E), voltage (Fig. 2G-H; Fig. 2—figure supplement 1B, C) and exhibits a long-lasting inhibitory effect (Fig. 2I, K). This sustained inhibitory effect of zinc can be partially restored by EGTA, a divalent ion chelator with high affinity for zinc (Fig. 2J, K; Fig. 2—figure supplement 1D, E). Previous studies indicate that zinc also regulates the Slo channel family, albeit with varying effects among Slo3 channel subtypes: zinc activates Slo1 channels (Hou et al.,

2010) while inhibiting Slo2.2 channels (Budelli et al., 2016; J. Zhang et al., 2023). Zinc may produce these opposite effects on Slo2.2 and Slo1 by binding to distinct, non-conserved sites among the subtypes (J. Zhang et al., 2023).

Zinc inhibition on mSlo3 channels is characterized by the slow dissociation of zinc from the channel (Fig. 2I–K). Similar slow dissociation has been observed in zinc-regulated ion channels and receptors, such as NMDA and glycine receptors (Paoletti et al., 1997; Trombley et al., 2011). In NMDA receptors, high zinc affinity results in slow dissociation, which has been suggested to have a physiological relevance in neurons (Paoletti et al., 1997).

Slow dissociation typically indicates high-affinity binding. In contrast to NMDA receptors, which exhibit IC_{50} values ranging from nM to μ M, mSlo3 has a relatively high IC_{50} for zinc inhibition (145.2 μ M). This might be due to the ~40-second zinc application time in our study, which may have been insufficient to fully observe the effects at lower concentrations (Fig. 2B, C). Extended exposure of intracellular zinc in sperm could potentially result in greater inhibition.

Furthermore, our results also show that when mSlo3 is coexpressed with Lrrc52 the lower concentration of zinc was sufficient to inhibit mSlo3 channels (Fig. 3). Although we still did not identify the exact mechanism underlying it, Lrrc52 shifts the gating of mSlo3 to more hyperpolarized potentials and lower pH, making the channel easier to open (Leonetti et al., 2012; Yang et al., 2011; X. H. Zeng et al., 2015). Considering that the zinc inhibition of Slo3 depends on membrane potential and intracellular pH, it is understandable that the presence of this auxiliary subunit would lead to more efficient inhibition by zinc.

Structural determinants of zinc inhibition in the mSlo3 channel

We identified potential key amino acids in the mSlo3 channel that may serve as zinc binding sites using MD simulations and amino acid mutagenesis (Fig. 4, 5; Fig. 4—figure supplement 1, Fig. 5—figure supplement 1). Previous studies on the Slo channel family indicates that zinc binding sites are located in RCK1 domain for Slo1 (Hou et al., 2010) and RCK2 domain for Slo2.2 (J. Zhang et al., 2023). However, in mSlo3 channel, our results didn't show that either RCK1 or RCK2 may serve as zinc binding site (Fig. 4—figure supplement 1).

On the other hand, our analysis of current recovery for mSlo3 mutants in the VSD region revealed that E169A and E205A showed significant recovery upon wash-out with pH 8.0. This result aligns with MD simulations showing faster zinc dissociation for both E169A and E205A, indicating their contributions in zinc-mediated inhibition. Thus, the cluster of negative residues within the VSD serve as the zinc binding site in the mSlo3 channel.

Based on sequence alignment, E169 (mSlo3 numbering) is conserved among Slo family channels in mice, whereas E205 (mSlo3 numbering) is not (Fig. 5—figure supplement 1F). To date, no studies have examined the corresponding residues to E169 (E191 in mSlo1 or E176 in mSlo2.2) for their potential zinc sensitivity, likely because the established zinc binding sites in these channels are located in the RCK1 domain for Slo1 (Hou et al., 2010) and the RCK2 domain for Slo2.2 (J. Zhang et al., 2023). The identified zinc binding site in Slo2.2 is conserved in Slo2.1 but is absent in both Slo1 and Slo3 (J. Zhang et al., 2023), further suggesting that zinc regulation differs among Slo family members. Although regions surrounding E191 or E176 may still provide additional insights into zinc regulation and could be of interest for future investigation, E205 stands out because, unlike E169, it is not conserved across the Slo family, making it unique to mSlo3 and potentially a specific determinant of zinc sensitivity in this channel.

However, our single amino acid mutations in this site, which contain clustered negative residues, did not significantly change zinc-mediated current reduction compared to the wildtype. This may be due to mutating one single amino acid might not be sufficient to fully elucidate other contributing residues in the predicted mSlo3 zinc binding site. Thus, more extensive mutagenesis studies will be needed to fully understand zinc inhibition in mSlo3.

It is worth noting that the incomplete loss of zinc sensitivity in these mutants suggests that additional mechanisms may participate in zinc modulation of Slo3. These may include modulation of nearby charged residues, structural rearrangements influenced by zinc binding, or the presence

of multiple zinc binding sites. Comparisons with Slo2.2 (J. Zhang et al., 2023 [DOI](#)), KCNQ4 (Gao et al., 2017 [DOI](#)), and voltage-gated calcium channels (Sun et al., 2007 [DOI](#)) further support the possibility of diverse molecular determinants for zinc inhibition. Our VCF, mutagenesis, and simulation data together indicate that zinc influences voltage sensor movement in mSlo3, which may suggest a distinct inhibitory mechanism that warrants further investigation.

Physiological relevance of zinc inhibition of the mSlo3 channel in mouse sperm

Zinc is thought to be involved in capacitation and the acrosome reaction, suggesting its role in regulating sperm fertilizing ability *in vitro* (Aonuma et al., 1978 [DOI](#)). This finding is supported by studies indicating that seminal zinc inhibits sperm's ability to undergo the acrosome reaction (Riffo et al., 1992 [DOI](#)). However, the specific function of intracellular zinc in relation to the capacitation process has not been thoroughly investigated. Previous research indicates that zinc inhibits sperm motility in ejaculation medium as well as acrosome reaction (Henkel et al., 1999 [DOI](#); Riffo et al., 1992 [DOI](#)), leading to the hypothesis that zinc might influence capacitation.

Slo3 plays a critical role in sperm fertility, since mice lacking Slo3 are infertile despite their normal sperm morphology (Santi et al., 2010 [DOI](#)). Furthermore, our previous work showed that VSP-deficient sperm exhibited enhanced Slo3 channel activity, which indirectly increases Ca^{2+} influx and results in abnormal sperm motility (Kawai et al., 2019 [DOI](#)). Taken together, our findings suggest that intracellular zinc is potentially a key regulator of sperm capacitation, functioning through inhibition of the Slo3 channel. Although these results support a mechanistic link between zinc and Slo3 activity, future studies that combine genetic or pharmacological modulation of Slo3 with comprehensive capacitation analyses will be required to define its physiological impact in more detail. Within this context, this study highlights the potential importance of intracellular zinc in the regulation of sperm capacitation.

In addition to that, to date, there are only few reports on the effect of zinc on other sperm ion channels, and none have been reported in mouse sperm. One important study was reported by (Jeschke et al., 2021 [DOI](#)), in which seminal zinc was found to inhibit prostaglandin-induced activation of CatSper, a sperm-specific Ca^{2+} channel, in human sperm. The complex opposing action of seminal zinc and prostaglandins on CatSper may help preventing premature activation of CatSper in the ejaculate and act as a dilution sensor, although this study does not provide direct evidence for zinc acting directly on CatSper (Jeschke et al., 2021 [DOI](#)).

Taken together, this study provides new insights into the zinc-dependent modulation of sperm ion channels, underscoring the central role of Slo3 and offering a foundation for future exploration of the complex molecular mechanisms governing fertilization.

Materials and Methods

Ethical approval

All animal experiments were approved by the Animal Care and Use Committee of the Osaka University Graduate School of Medicine and performed in accordance with its guidelines.

Drug and Reagent

Zinc pyrithione was purchased from Fujifilm Wako (Osaka, Japan). TPEN was obtained from Cayman (Ann Arbor, MI, USA). Gramicidin was obtained from Merck (Kenilworth, NJ).

Patch Clamp Recordings from Spermatozoa

Patch clamp recordings from spermatozoa were conducted as previously described (Kawai et al., 2019 [DOI](#); Kirichok et al., 2006 [DOI](#)). Briefly, sperm were harvested from the mouse corpus epididymis using an HS-based solution containing (in mM): 135 NaCl, 5 KCl, 2 CaCl_2 , 1 MgSO_4 , 20 HEPES, 5

glucose, 10 lactic acid, and 1 sodium pyruvate (pH 7.4). After 10 minutes, the supernatant was collected, centrifuged, washed twice with the HS-based solution, resuspended, and placed on untreated glass coverslips.

For conventional whole-cell recordings, the intracellular solution contained (in mM): 120 KCl, 3 MgCl₂, 40 MES, and 0.3 EGTA (pH 6.0), with the osmolality adjusted to approximately 300 mOsm/kg using sucrose. The same HS-based solution was used as the bath solution.

For perforated patch recordings, the intracellular solution included (in mM): 120 KCl, 3 MgCl₂, 40 HEPES, 0.3 EGTA (pH 7.0), and 0.05 mg/ml gramicidin. HS-based solution was used as the extracellular solution. Once pores formed, the access resistance ranged between 50 and 100 MΩ.

Recording pipettes were made with borosilicate glass (BF-150-86-10; Sutter Instruments, CA, USA) using a puller (P-97; Sutter Instruments). The recordings were conducted using an Axopatch 200B amplifier (Molecular Devices, CA, USA) and sampled at 5 kHz using a Digidata 1550A system (Molecular Devices) with pCLAMP 10.5 software (Molecular Devices). Low pass filter of 20 Hz was applied to increase S/N ratio.

Zinc imaging from Spermatozoa

Sperm were isolated from the cauda epididymis of mice and transferred to HS medium. The sperm were then loaded with 10 μM FluoZin3-AM (Invitrogen) and 0.05% Pluronic F-127 in HS medium, followed by a 30-minutes incubation at 37°C. After incubation, the sperm were washed with HS medium and incubated on Cell-Tak (Corning, Bedford, MA)-coated round cover glasses for 10 minutes. The cover glass was mounted on a perfusion chamber for imaging experiments.

Fluorescence signals were detected using a filter set (Ex 495/10; DM 515; Em 535/25) with an IX-81 upright microscope (Olympus, Tokyo, Japan) equipped with a C9100 EMCCD camera (Hamamatsu Photonics, Shizuoka, Japan) and Metamorph imaging software (Molecular Devices). Images were captured every 30 seconds. Fluorescence intensity is measured by the signal taken from the whole head and the proximal part of tail in sperm.

For capacitation, TYH medium (Toyoda et al., 1971 [link](#)) was used, consisting of: 120 mM NaCl, 4.8 mM KCl, 1.2 mM KH₂PO₄, 5.6 mM glucose, 1.0 mM sodium pyruvate, 1.7 mM CaCl₂, 1.2 mM MgSO₄, 25 mM NaHCO₃, 4.0 g/L ALBMAX I (Thermo Fisher Scientific), and penicillin (50 units/ml)-streptomycin (50 μg/ml).

The experiment focused on the head and tail regions of spermatozoa. For analysis, cells in which both the head and tail were visible throughout the entire experiment were selected.

Sperm motility analysis

Sperm velocity was analyzed as previously described (Miyata et al., 2021 [link](#)). Briefly, spermatozoa isolated from the cauda epididymis were suspended in TYH medium. The TYH medium contained: 120 mM NaCl, 4.8 mM KCl, 1.2 mM KH₂PO₄, 5.6 mM glucose, 1.0 mM sodium pyruvate, 1.7 mM CaCl₂, 1.2 mM MgSO₄, 25 mM NaHCO₃, 4.0 g/L ALBMAX I (Thermo Fisher Scientific), penicillin (50 units/ml) - streptomycin (50 μg/ml), and 0.6% Phenol Red.

Average path velocity (VAP), curvilinear velocity (VCL), straight-line velocity (VSL), straightness (STR), and linearity (LIN) was measured using the CEROS II sperm analysis system (Hamilton Thorne Biosciences, MA, USA) at 10 minutes and 2 hours after incubation. In some experiments, 20 μM TPEN was directly added to the TYH medium.

Molecular Biology

All Slo3 mutants were constructed from WT *mus musculus* Slo3 (mSlo3) in PSD64TF. Single amino acid mutations were introduced using PrimeSTAR Mutagenesis Basal Kit (Takara Bio Inc.). Chimeras were generated using In-Fusion HD Cloning Kit (Takara Bio Inc.). The introduced mutations and generated chimeras were confirmed by DNA sequencing. The mMMESSAGE SP6 RNA transcription kit (Thermo Fisher Scientific) was used to transcribe WT, mutants, and chimeras of mSlo3 cRNAs from plasmid cDNA linearized by Xba1 restriction enzyme (Takara Bio Inc.).

Preparation of *Xenopus laevis* oocytes

As an anesthetic agent, 0.15% tricaine (Sigma-Aldrich / Tokyo Chemical Industry) was used for *Xenopus laevis* before surgical operation for isolation of oocytes. Follicular membranes were removed from isolated oocytes by collagenase treatment (2 mg ml⁻¹; type 1; Sigma-Aldrich or Collagenase-P from Roche) for 3 hr. Oocytes were then rinsed and stored in ND96 solution (96 mM NaCl, 2 mM KCl, 1.8 mM CaCl₂, 1 mM MgCl₂, and 5 mM 2-[4-(2-Hydroxyethyl)-1-piperazinyl]ethanesulfonic acid (HEPES), pH 7.5, with NaOH) supplemented with 0.1 mg/mL gentamycin (FUJIFILM Wako Pure Chemical Corporation or Nacalai Tesque, Inc.) and 5 mM sodium pyruvate. The incubation temperature was 18 °C.

Expression of mSlo3 channel

Xenopus oocytes injected with 15 ng of WT mSlo3 cRNA and incubated for 2 days at 18°C. For mSlo3 mutants, oocytes were injected with 50 ng–60 ng of cRNA and incubated for 1–3 days, depending on the desired expression level for patch-clamp and voltage-clamp fluorometry (VCF) experiments. cRNA for mLrrc52 was also injected for data shown in Fig. 3 [Fig. 3—figure supplement 1](#) [Fig. 6](#) [Fig. 6—figure supplement 1F-H](#).

Oocyte patch-clamp recording of mSlo3

Patch-clamp recordings of *Xenopus* oocytes were performed using an Axopatch-200B amplifier (Molecular Devices) connected through an AD/DA converter (Digidata 1440A; Molecular Devices) and operated with pClamp software. Current traces were filtered at 5 kHz and sampled at 100 kHz. Vitelline membrane was manually removed using forceps after incubation for 1 to 5-minute in a hypertonic solution containing 192 mM NMDG, 4 mM KCl, 3.6 mM CaCl₂, 2 mM MgCl₂, and 10 mM HEPES, pH 7.4, with HCl. Leak currents were subtracted by using a P/-8 protocol for step pulses recordings only. Patch pipettes, with resistance ranging from 1 to 3 MΩ were pulled from borosilicate glass (Drummond Scientific Company). The pipette solution contains 0.5 mM potassium gluconate, 0.5 mM KCl, 1.1 mM KOH, 10 mM HEPES, 159 mM sodium gluconate, and 2 mM MgCl₂, pH 7.1, adjusted with methanesulfonic acid.

Patches were perfused with: (1) bath solution containing 184 mM potassium gluconate, 10 mM KOH and 10 mM HEPES in pH 6.0, 7.5, and 8.0; (2) zinc-containing bath solution at pH 7.5 and pH 8.0, with zinc concentrations of 0.1, 1, 10, 100, and 1000 μM. (3) bath solution containing 5 mM EGTA in pH 8.0. All pH values were adjusted using methanesulfonic acid. Slo3 macroscopic currents were evoked by an alkaline pH and by a depolarizing ramp pulse (from -100 mV to +100 mV with the holding potential of either -60 mV or 0 mV) or step pulses (from -100 mV to +180 mV with the holding potential at -60 mV). All bath solutions were perfused using a high-speed perfusion system ALA-VM8 OCTAFLow (ALA Scientific Instruments Inc.)

Voltage-clamp fluorometry (VCF) recording of mSlo3

In mSlo3 VCF construct, the target labeling site (in S3-S4 linker) was substituted with cysteine residue. Two native, extracellular cysteines were mutated to avoid non-specific labeling (mSlo3 C19S/C131S). Oocytes for VCF recordings were labeled with Tetramethylrhodamine-5-maleimide (TMRM) (Invitrogen). In prior to VCF recordings, oocytes were incubated in a labeling solutions (120 mM K-methanesulfonate (MES), 2 mM CaCl₂, and 10 mM HEPES pH 7.4) containing 10 μM TMRM (Savalli et al., 2006 [Fig. 3](#)) on ice for 1 hour. Labeled oocytes were then rinsed in TMRM-free ND96 solution four times and kept in the dark until and throughout the recording.

Oocytes were placed in a recording chamber with the animal pole facing upward and held at -60 mV. For screening experiments, step pulses ranging from -80 mV to +160 mV were applied, while pulses from -160 mV to +160 mV were used for direct zinc injection experiments. For the VCF recording in the absence and presence of zinc, voltage steps were first applied to record the initial response. Subsequently, a puff of 10 mM zinc solution was manually injected into the same oocyte using a glass needle and positive pressure. Sixty seconds post-injection, a second set of step pulses was applied to record the response under zinc application. In VCF recording, borosilicate glass

capillaries (Harvard Apparatus) were used with a resistance of 0.2–0.4 M Ω when filled with 3 M KOAc and 10 mM KCl. Bath solution contained 96 mM NaCl, 2 mM KCl, 1.8 mM CaCl₂, 1 mM MgCl₂, and 5 mM HEPES, pH 7.4–7.5, with NaOH was used for recordings.

Fluorometric recordings were performed with an upright fluorescence microscope (Olympus BX50WI) equipped with a water immersion objective lens (Olympus XLUMPLAN FL 20x/1.00) to collect the emission light from the voltage-clamped oocytes.

The light from LED source (MCWHL8 Thorlabs) was applied through a band-pass excitation filter (531 nm/40 nm BrightLine: Semrock). Emitted light was passed through band pass emission filter of 593nm/40nm (BrightLine, Semrock). The emission signals were detected by a photomultiplier (H10722-20; Hamamatsu Photonics). The detected emission intensities were acquired by a Digidata 1440A (Axon Instruments) and Clampex 10.3 software (Molecular Devices) at 25 kHz for VCF mutants screening and 50 kHz for VCF recording in the absence and presence of zinc. To improve the signal-to-noise ratio, VCF recording was repeated 3 times. Averaged data were used for data presentation and analysis.

Three-dimensional structural modeling of mouse Slo3 and prediction of zinc binding site

The amino acid sequence of mouse Slo3 (mSlo3) (O54982-1) was obtained from the UniProt database (Fig. 4 [↗](#)) and was retrieved from NCBI NP_032458.3 (Fig. 5 [↗](#)). The sequence (only the transmembrane region from residue M1 to G330, or the full-length protein) was input to the AlphaFold3 (Abramson et al., 2024 [↗](#)) structural modelling web platform to generate an initial structure with (Fig. 4A [↗](#)) or without (Fig. 5A [↗](#)) 50 zinc ions. All the structural data presented in this study were generated using PyMOL molecular graphics system ver. 2.5.0 (Schrodinger LLC).

Flooding simulations

The full-length mSlo3 from the AlphaFold3 webserver was trimmed to remove long disordered regions (residue 1–18, 616–698 and 1050–1121), resulting in 12 separate chains. The mSlo3 complex was embedded into the POPC bilayer and solvated in 0.15 M NaCl and 0.15 ZnCl₂ using CHARMM-GUI Membrane Builder (Wu et al., 2014 [↗](#)). The production run simulations were simulated with a 2 fs timestep using the CHARMM36m forcefield (Huang et al., 2016 [↗](#)) in GROMACS 2023.3 (Abraham et al., 2023 [↗](#)). The system was then energy minimized and equilibrated using the standard six steps CHARMM-GUI equilibration protocol. This includes the following set-up: The protein backbone was restrained at the force constant of 4000, 2000, 1000, 500, 200 and 50 kJmol^{−1}nm^{−2}, the protein side chains were restrained with the force constant of 2000, 1000, 500, 200, 50, and 0 kJmol^{−1}nm^{−2}, the lipids non-H atoms were restrained at the force constant of 1000, 400, 400, 200, 40 and 0 kJmol^{−1}nm^{−2} and the dihedral restraint was set at the force constant of 1000, 400, 200, 200, 100 and 0 kJmol^{−1}rad^{−2}. The simulations were equilibrated with a 1 fs timestep for 125 ps for the first three steps, and then to a 2 fs timestep for 500 ps the next two, and 5 ns for the final step. The first two steps were conducted with the NVT ensemble, whereas the last four were conducted with the NPT ensemble. All equilibration runs were conducted at 310 K using Berendsen thermostat (Berendsen et al., 1984 [↗](#)). In all NPT ensemble equilibration, the semi-isotropic pressure was maintained at 1 bar using Berendsen barostat. The production runs were conducted for 200 ns under 310 K using the v-rescale thermostat (Bussi et al., 2007 [↗](#)). The semi-isotropic pressure of all systems was maintained at 1 bar using a C-rescale barostat (Bernetti & Bussi, 2020 [↗](#)). All simulations were carried out in triplicates, where velocity was randomly generated at the beginning of every simulation. All simulation frames were saved every 0.1 ns.

Zn stability

Based on the flooding simulations, the binding pose for Zn²⁺ is adjusted so that the ion is 3–4 Å away from E169. This results in the system with 4 Zn²⁺ per system, one per subunit. The system was then energy minimized using the steepest descent algorithm and equilibrated with the Ca

restraint at 1000 kJ kJmol⁻¹nm⁻² for 10 ns. The production run was conducted for 200 ns, with the same barostat and thermostat setting as the flooding simulation.

Data Analysis

Inside-out patch-clamp data were analyzed using Clampfit 11.3 software (Molecular Devices), Igor Pro 6.22A (Wavemetrics), and GraphPad Prism 10. Analysis of current-voltage (I-V) were obtained from the maximum current amplitude at steady-state. Normalization for figure 2H [\[1\]](#) was performed by dividing the absolute current of mSlo3 at pH 8.0 of each voltage by the absolute current at the pre-determined highest voltage that still produced a stable mSlo3 current (i.e., good patch, good clamp). In this analysis, +140 mV was chosen as the highest voltage for normalization, since in few cells the patch was lost at +160mV and +180mV. Similar to the control condition, the absolute current of mSlo3 in the presence of 100 μM zinc was normalized to the absolute current of the control at +140 mV. I-V relationship was fitted with Boltzmann equation:

$$I(V) = \frac{G}{1 + \exp\left(\frac{V_{50}-V}{s}\right)} \times (V - V_{\text{rev}}) \quad (1)$$

where G is a scaling factor (maximum conductance or current), V₅₀ is the midpoint, and s is the slope value.

And then the I-V relationship in the presence of 100 μM of ZnCl₂ was fitted with Woodhull equation (Woodhull, 1973 [\[2\]](#)) to express the current as a function of voltage, incorporating both the voltage-dependent gating of the channel and the voltage-dependent blocking of the channel:

$$I(V) = \frac{G}{1 + \exp\left(\frac{V_{50}-V}{s}\right)} \times (V - V_{\text{rev}}) \times \frac{1}{1 + \exp\left(\frac{V_{\text{block}}-V}{\text{Slope}_{\text{block}}}\right)} \quad (2)$$

$$\text{Slope}_{\text{block}} = \frac{RT}{-\delta zF} \quad (3)$$

where V_{block} is the voltage at which blocking is half-maximal, Slope_{block} is the slope factor for the voltage dependence of the blocking effect, and δ is the fraction of the membrane potential that the blocking ion experiences.

V_{rev} is the reversal potential and was calculated with following equation as described in (Schreiber et al., 1998 [\[3\]](#)) using Goldman-Hodgkin-Katz equation:

$$E_{\text{rev}} = \frac{RT}{F} \ln \frac{[K^+]_{\text{out}} + P[Na^+]_{\text{out}}}{[K^+]_{\text{in}} + P[Na^+]_{\text{in}}} \quad (4)$$

where P is the Na⁺ to K⁺ permeability ratio (Schreiber et al., 1998 [\[3\]](#)).

Conductance-voltage (G-V) relationship of Slo3 were obtained from conductance calculation using the following equation:

$$G(V) = \frac{I}{V - V_{\text{rev}}} \quad (5)$$

V_{rev} was calculated using equation (4) and fitted to Boltzmann equation in (1), adjusted for G-V.

IC₅₀ (half maximal inhibitory concentration) was calculated using equation while constraining the Hill Slope to a constant value of -1.0:

$$Y = I_{\min} + \frac{I_{\max} - I_{\min}}{1 + \frac{IC_{50}}{X}} \quad (6)$$

where $I_{\max}$ is the maximal current when inhibition is minimal, $I_{\min}$ is the minimal current inhibition is maximal, and X is the inhibitor concentration.

Arithmetic operations were performed by Clampfit to compensate bleaching from the fluorescence traces:

$$[\text{Compensated trace}] = [\text{Recorded Ftrace}] \times (1 - (R \times [\text{time}])) \quad (7)$$

where [time] is the value of the point given by Clampfit.

All the compensated traces were then normalized by setting each baseline level (F signal at -60 mV, the holding potential) to be 1, to calculate the % F change:

$$(\Delta F/F; \Delta F = F + 160\text{mv}^2 - F_{\text{baseline}}; F = F_{\text{baseline}})$$

Conductance-voltage (G-V) relationship for VCF data were obtained from conductance calculation using equation (5). Fluorescence-voltage (F-V) relationship were obtained from the normalized $\Delta F/F$ and fitted with the equation (1).

Statistical analysis

Statistical analysis was performed by either one-way ANOVA, unpaired t-test or paired t-test. Following one-way ANOVA, Dunnett's post-hoc test was applied. The data were expressed as mean $\pm$ s.e.m. with n indicating the number of samples. Values of p were stated in each bar graph. All the statistical analysis and the bar graphs were performed and generated with GraphPad Prism 10.

Figure supplements

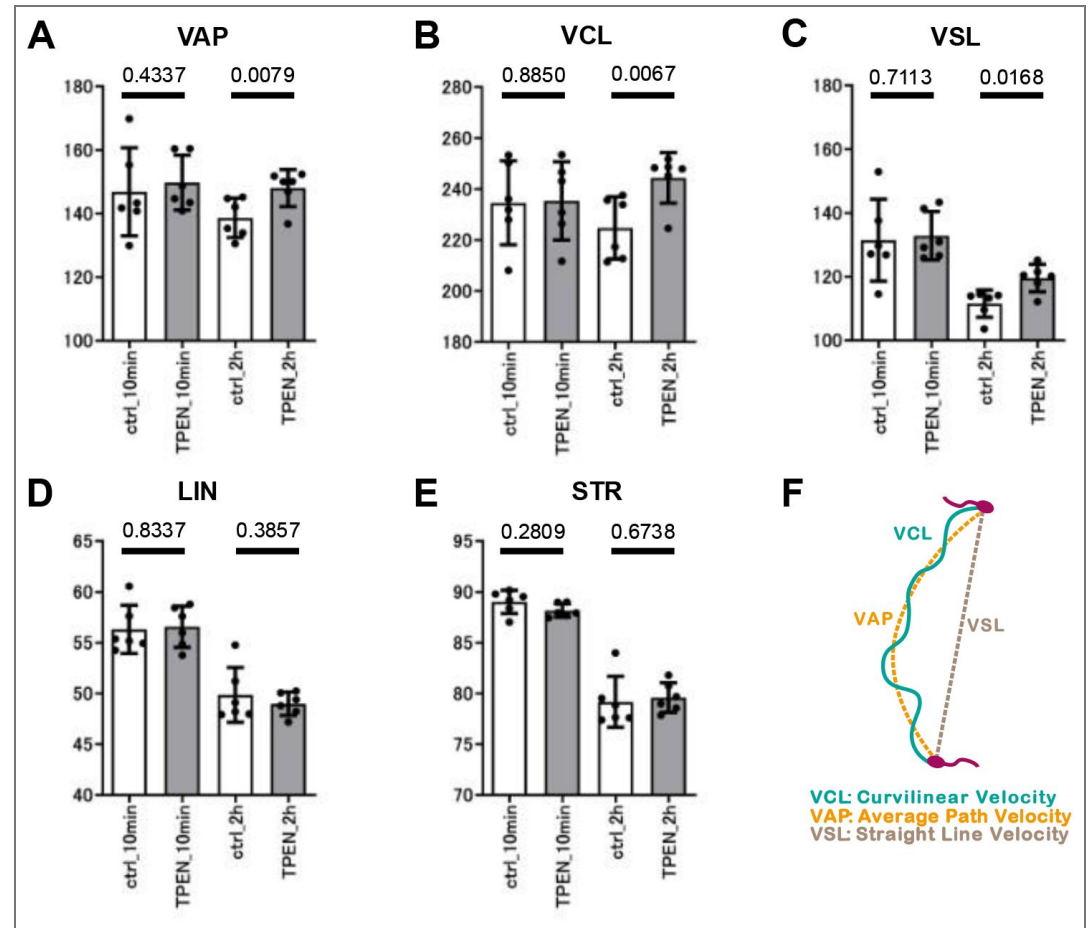


Figure S1—figure supplement 1. Effects of intracellular zinc on sperm motility before and after capacitation. Comparison of VAP, VCL and VSL show that intracellular zinc affects these kinds of motions only after the sperm underwent capacitation. (**A-E**) Quantitation of sperm motility parameters at: 10 min (control); 10 min in the presence of zinc chelator TPEN; 2 h after TYH incubation, a well-established capacitation-inducing medium (control); and 2 h after THY incubation in the presence of TPEN. Bar graph shows the comparison of (**A**) VAP (average path velocity); (**B**) VCL (curvilinear velocity); (**C**) VSL (straight-line velocity); (**D**) STR (straightness); and (**E**) LIN (linearity) as depicted in (**F**). All error bars are $\pm$ SD centered on the mean.

Figure S1—figure supplement 2. The effects of zinc on Slo3 current in sperm cells.

(A) Comparison of normalized Slo3 current at +100 mV before NH_4Cl application between control (50.67 ± 20.67 pA/pF; $n=6$) and in the presence of $100 \mu\text{M}$ ZnPy (59.95 ± 30.24 pA/pF; $n=4$), $p=0.6156$, Welch's t-test. (B) Comparison of normalized Slo3 current at +100 mV after NH_4Cl application between control (144.8 ± 40.36 pA/pF) and in the presence of $100 \mu\text{M}$ ZnPy (77.59 ± 44.97 pA/pF; $n=4$), $p=0.0684$, Welch's t-test. All error bars are $\pm$ SD, centered on the mean.

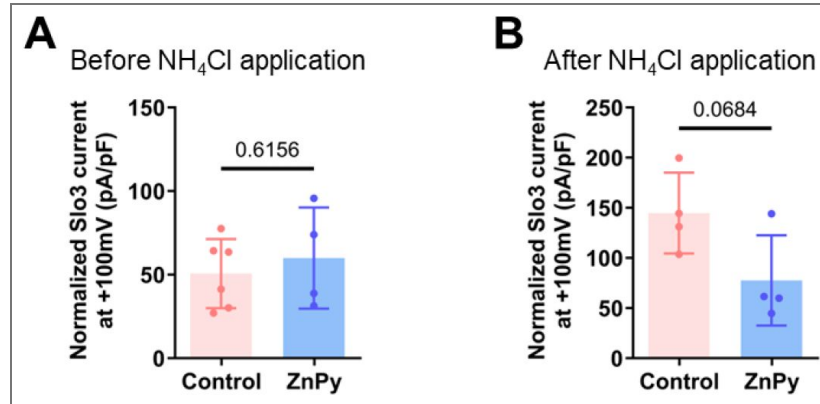
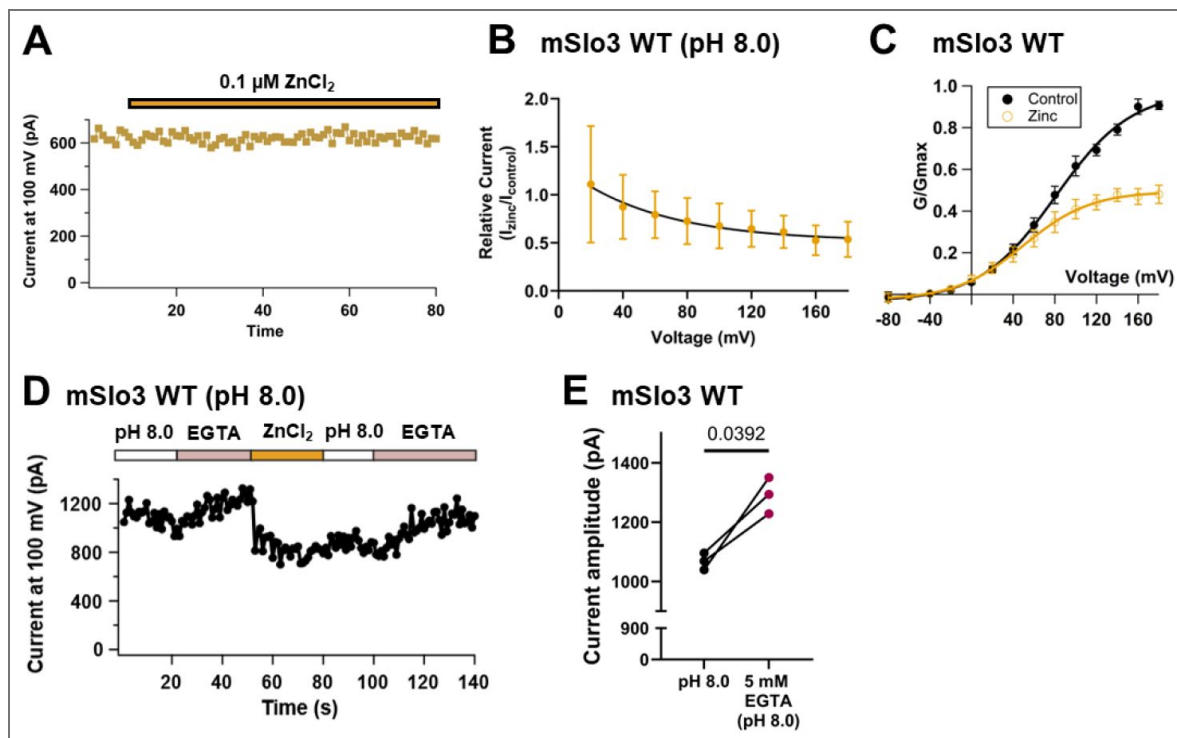


Figure S2—figure supplement 1. Zinc inhibits mSlo3 current.

(A) Time course of Slo3 current at +100 mV upon the application of $0.1 \mu\text{M}$ zinc. (B) Relationship between relative current ($I_{\text{zinc}}/I_{\text{control}}$) with membrane voltage at pH 8.0 shows that zinc inhibits mSlo3 current in voltage-dependent manner. (C) G-V relationship comparison of mSlo3 current between control (pH 8.0; black filled circle) and upon the application of $100 \mu\text{M}$ zinc in pH 8.0 (yellow orange circle) (V_{50} control = 80.67 ± 2.32 ; V_{50} zinc = 49.677 ± 1.38 ; $n=13$). Normalization was done based on the maximum conductance in control. (D) Time course of the change in current amplitude at +100 mV in response to pH 8.0, 5 mM EGTA in pH 8.0, $100 \mu\text{M}$ zinc, pH 8.0, and 5 mM EGTA in pH 8.0 as indicated (white: pH 8.0; yellow orange: $100 \mu\text{M}$ zinc in pH 8.0; pale lilac: 5 mM EGTA in pH 8.0). The baseline current showed a slight increase upon perfusing with 5 mM EGTA, indicating that endogenous zinc is already present in mSlo3, and it is affecting the baseline current. (E) Comparison of percentage of mSlo3 current evoked by pH 8.0 and 5 mM EGTA in pH 8.0 ($p=0.0392$). Statistical analysis was done by paired t-test, $n=3$. All error bars are $\pm$ SD centered on the mean.



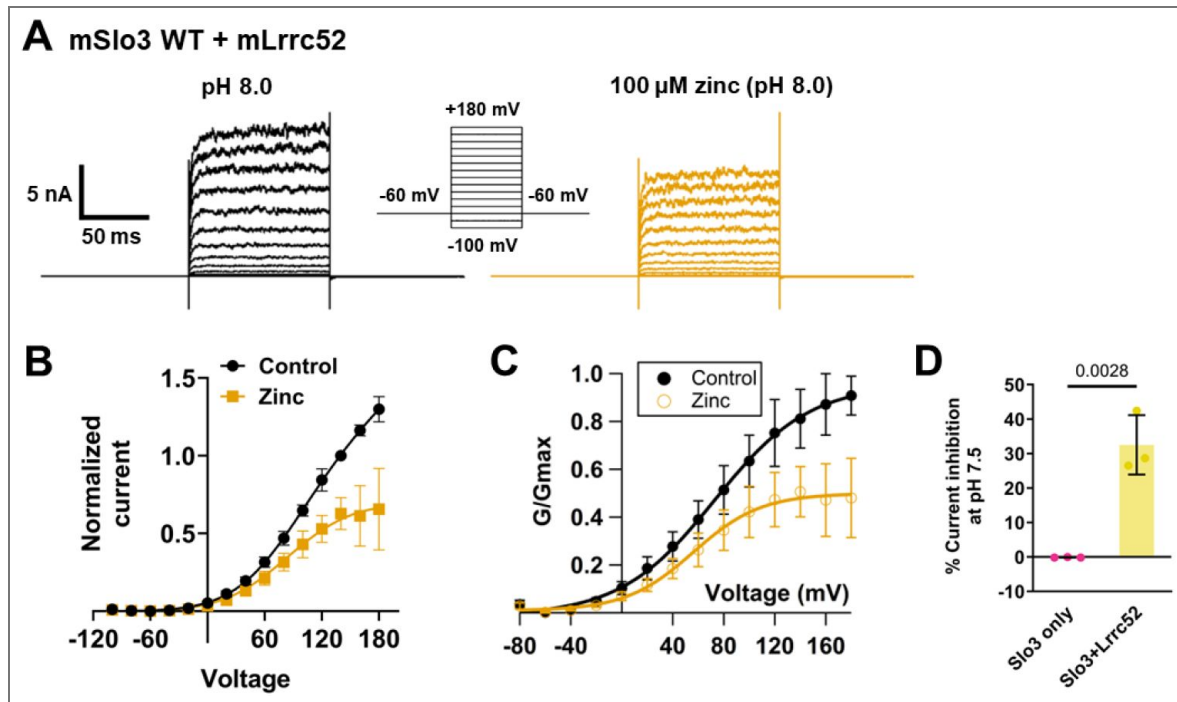


Figure S3—figure supplement 1. Zinc inhibits mSlo3 current when coexpressed with auxiliary Lrrc52.

(A–C) Inside-out patch-clamp recording of mSlo3 co-expressed with mLrrc52 with $pH_i=8.0$ by applying step pulses from -100 mV to $+180$ mV with the holding potential of -60 mV. Zinc inhibition is voltage-dependent. (A) Representative current traces upon the application of pH 8.0 (black) and 100μ M of zinc in pH 8.0 (yellow orange). (B) I–V relationship of control (pH 8.0; black) and zinc (100μ M; yellow orange). I–V curve was fitted by Boltzmann equation for control and followed by Woodhull equation for zinc inhibition ($V_{50 \text{ Ctrl}} = 74.01$; $\text{Slope}_{\text{Ctrl}} = 36.47$; $V_{\text{block Zn}} = 185.5$; $\text{Slope}_{\text{block Zn}} = -130.4$; $n=8$). (C) G–V relationship comparison of mSlo3 current co-expressed with mLrrc52 between control (pH 8.0; black filled circle) and upon the application of 100μ M zinc in pH 8.0 (yellow orange circle) ($n = 13$). $V_{50 \text{ control}} = 72.93 \pm 2.06$ $V_{50 \text{ zinc}} = 56.22 \pm 2.75$. Normalization was done based on the maximum conductance in control. (D) Comparison of the percentage of Slo3 current inhibition at pH 7.5 between Slo3 only (-0.13 ± 0.08 %; $n=3$) and Slo3+Lrrc52 (32.51 ± 8.62 %; $n=3$), $p=0.0028$, unpaired t -test. All error bars are $\pm$ SD centered on the mean.

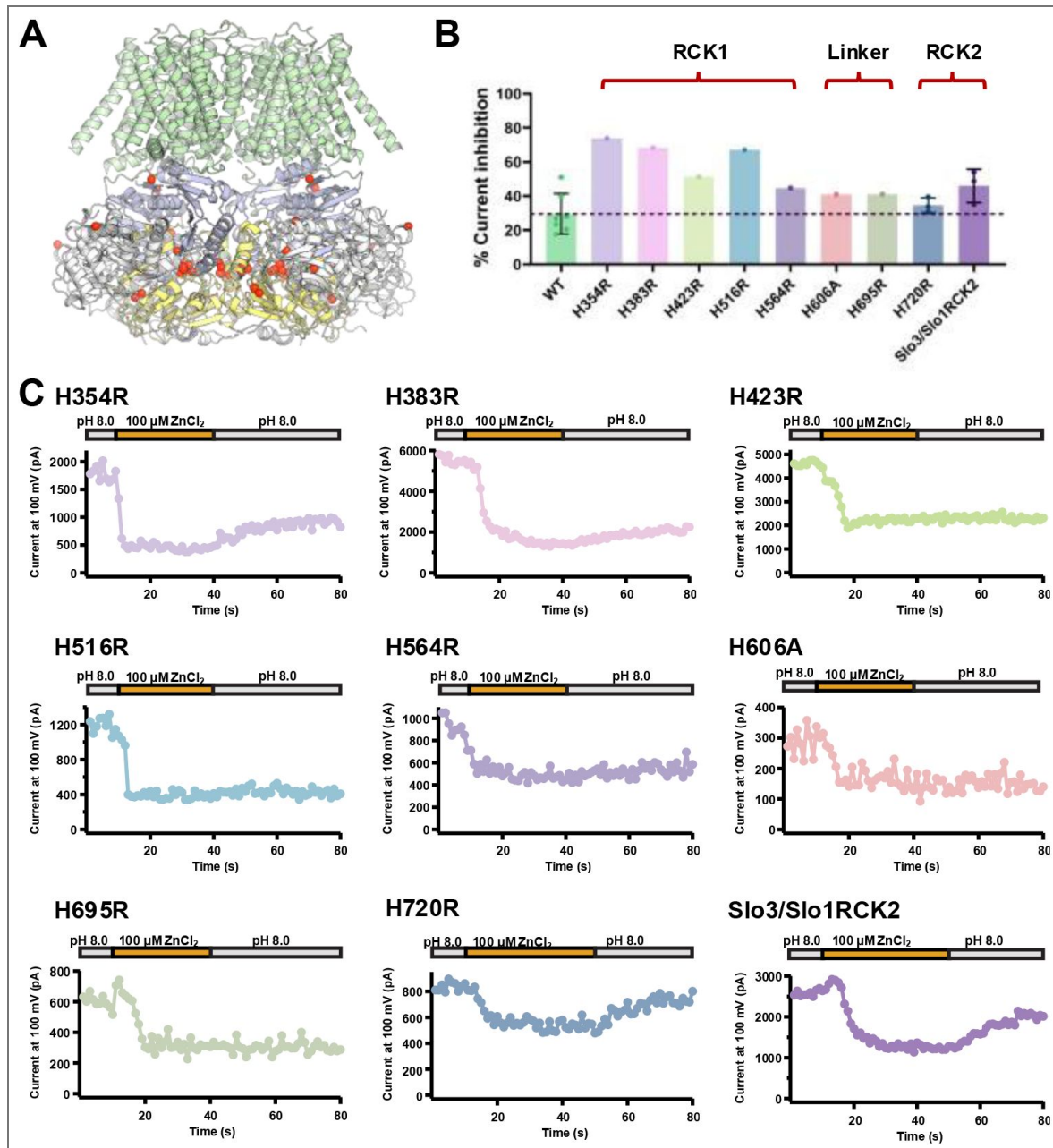


Figure S4—figure supplement 1. Computational studies and scanning mutagenesis of zinc binding site in mSlo3 channel.

(A) The full-length Slo3 tetramer structure predicted by AlphaFold3, with RCK1 and RCK2 represented in blue and yellow cartoons, respectively, and 50 Zn^{2+} ions depicted as red spheres. (B) Comparison of the percent current inhibition upon the application of 100 μM zinc between wildtype and all the histidine mutants scanned (histidine residues located in RCK1, RCK1/RCK2 linker, RCK2, and the chimera in which the RCK2 of mSlo3 is replaced with the RCK2 from mSlo1). Dotted black line indicates mean of control (WT). (C) Time course of the change in current amplitude at +100 mV in response to pH 8.0, 100 μM zinc, and pH 8.0 for wash-out as indicated for all the mutants as stated. All error bars are $\pm$ SD centered on the mean.

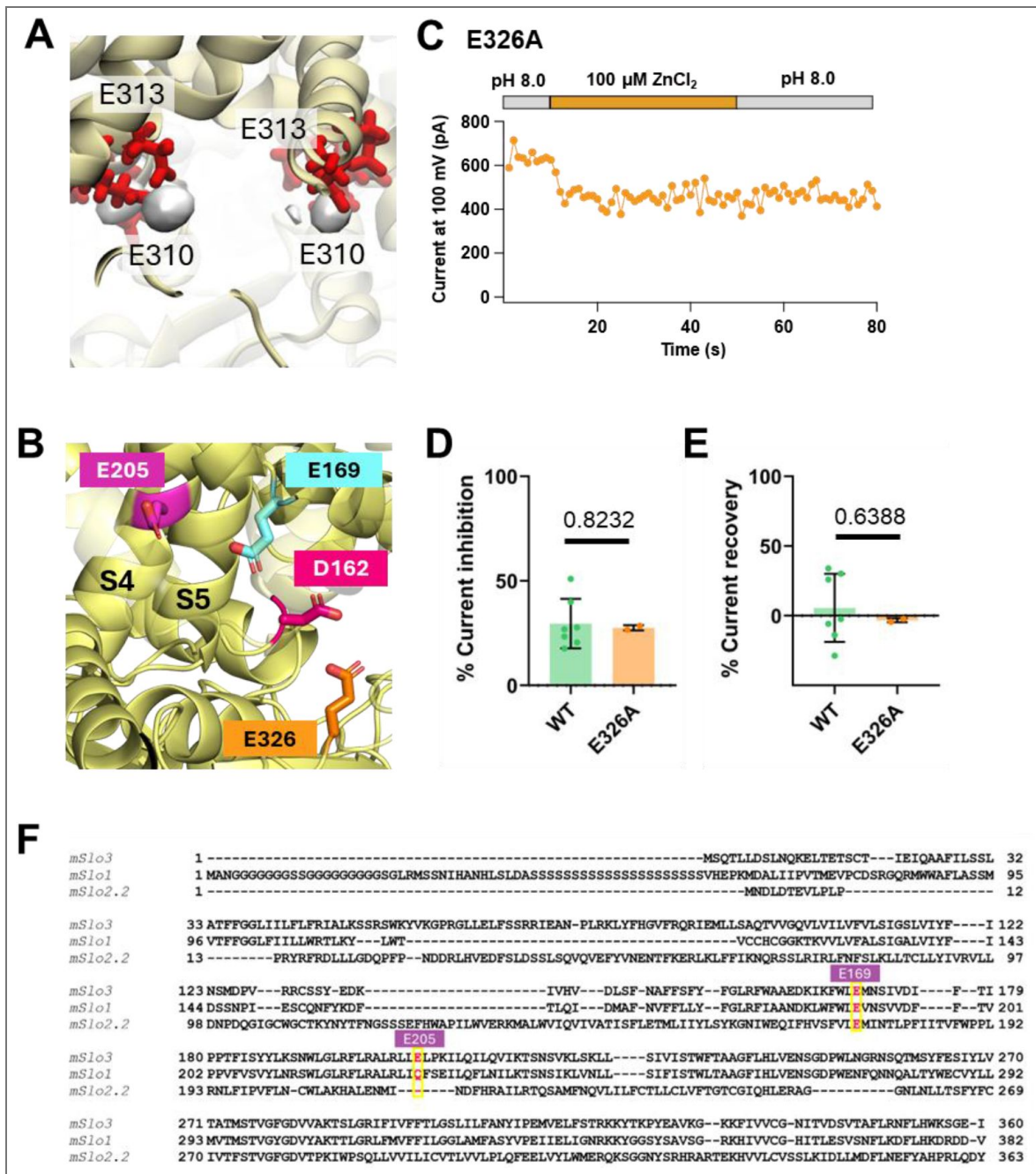


Figure S5—figure supplement 1. Computational studies and mutagenesis of zinc binding site in mSlo3 channel.

(A) The density of Zn ions within 4 Å of the protein molecules was calculated using VolMap averaged across 200 ns flooding simulations ($n = 3$). Key acidic contacting residues are shown as red sticks. (B) Tetrameric mSlo3 structure representation of selected region which indicates D162 (S2-S3 linker), E169 (S3 domain), E205 (S4 domain), and E326 (intracellular linker between TM and RCK1) which have been mutated to elucidate the zinc binding site in mSlo3. (C) Time course of the change in current amplitude at +100 mV in response to pH 8.0, 100 μ M zinc, and pH 8.0 for wash-out as indicated for E326A. (D) Comparison of the percent current inhibition upon the application of 100 μ M zinc between wildtype and E326A ($p=0.8232$, unpaired t-test, $n=2-7$). (E) Comparison of percentage of current recovery upon wash-out using pH 8.0 between WT and E326A ($p=0.6388$, unpaired t-test, $n=2-7$). All error bars are $\pm$ SD centered on the mean. (F) Sequence alignment of mSlo3 (NP_032458.3), mSlo1 (NP_001240293.1) and mSlo2.2 (NP_001138875.1) using ClustalO. E169 and E205 residues (mSlo3 numbering) are highlighted in red text with a yellow box.

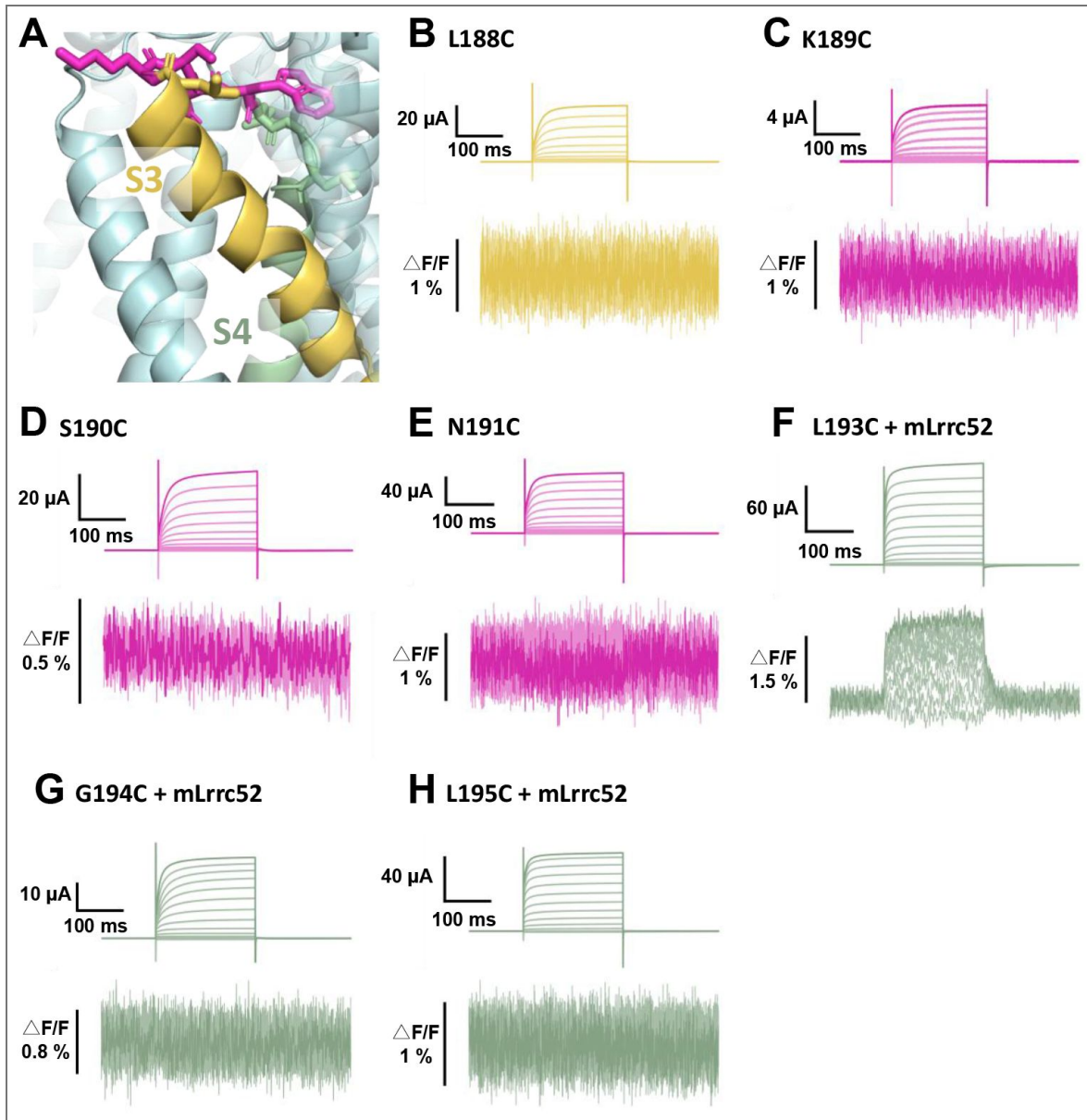


Figure S6—figure supplement 1. Scanning mutagenesis of mSlo3 channel VCF construct.

(A) Structural representation of the VCF scanning regions in the mSlo3 channel, highlighting the top of S3 (yellow), the S3-S4 linker (violet), and the top of S4 (mud green). (B-H) Representative current traces and fluorescence signals recorded upon voltage steps from -80 mV to +160 mV with a holding potential of -60 mV for L188C, K189C, S190C, N191C, L193C (+mLrrc52), G194C (+mLrrc52), and L195C (+mLrrc52), respectively (n=4-7; $\Delta F/F$ mSlo3 L193C+Lrrc52: = $1.15 \pm 0.48\%$ [n = 4]).

Data availability

Data sets are available upon request

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Peer reviews

Reviewer #1 (Public review):

Summary:

In their manuscript, Andriani et al. show intracellular zinc is exported from sperm during capacitation and suppresses the alkalinization-induced hyperpolarization in sperm. Intracellular zinc inhibits Slo3 current, which is enhanced by the co-expression of gamma

subunit Lrrc52. Computational studies reveal that the Zn binding site on mSlo3 is located near E169 and E205, which are involved in the sustained zinc inhibition of mSlo3 current. The authors propose that intracellular zinc play a key role of sperm capacitation by inhibiting the Slo3 channel.

Strengths:

Overall, the work appears well designed (e.g., oocyte patch-clamp experiments), and clearly presented. Three-dimensional structural modeling and flooding simulations are executed.

Weaknesses:

The simple mutagenesis analysis of E169 and E205 showed partial abolishment, but the molecular mechanism by which zinc inhibits Slo3 current is not yet fully shown. The authors should consider performing more extensive experiments, such as creating double mutants or combination mutants involving other residues. Additionally, could other mechanisms explain the role of zinc in regulating the Slo3 current?

While elucidating the mechanism of Slo3 is interesting, there is substantial literature indicating how zinc regulates channel functions at a molecular level. Given this, the manuscript should provide a deeper understanding by clearly elucidating the molecular mechanism of the regulation of Slo3 current by zinc.

The manuscript includes no experimental data on the mechanism of intracellular zinc export during sperm capacitation, despite being crucial for the regulation of sperm function.

<https://doi.org/10.7554/eLife.105450.2.sa2>

Reviewer #2 (Public review):

Summary:

In this paper, Andriani and colleagues are examining the potential role of Zn flux in sperm and its effect on Slo3 channels. This is an interesting question that is likely critical to how sperm function properly and Slo3 channels are a possible candidate for a downstream molecule that is impacted by Zn. In this paper the authors using Zn imaging, sperm motility assays, and electrophysiology to show that Zn flux has impacts on sperm function. They then go on to look at the impact Zn has on Slo3 current and propose a binding site based on MD simulations. Revisions of the paper added new critical controls and improved description of the methodology.

Strengths:

The question of how Zn flux impacts membrane potential and sperm motility is an important one. Moreover, Slo3 make present an interesting candidate or the target of Zn regulation. The combination of methods used here also has the potential to uncover mechanisms of Zn regulation of Slo3.

Weaknesses:

The responses sufficiently answered my original concerns.

<https://doi.org/10.7554/eLife.105450.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

In their manuscript, Andriani et al. show intracellular zinc is exported from sperm during capacitation and suppresses the alkalization-induced hyperpolarization in sperm. Intracellular zinc inhibits Slo3 current, which is enhanced by the co-expression of gamma subunit Lrrc52. Computational studies reveal that the Zn binding site on mSlo3 is located near E169 and E205, which are involved in the sustained zinc inhibition of mSlo3 current. The authors propose that intracellular zinc plays a key role in sperm capacitation by inhibiting the Slo3 channel.

Strengths:

Overall, the work appears well-designed (e.g., oocyte patch-clamp experiments), and clearly presented. Three-dimensional structural modeling and flooding simulations are executed.

Weaknesses:

The simple mutagenesis analysis of E169 and E205 showed partial abolishment, but the molecular mechanism by which zinc inhibits Slo3 current is not yet fully shown. The authors should consider performing more extensive experiments, such as creating double mutants or combination mutants involving other residues. Additionally, could other mechanisms explain the role of zinc in regulating the Slo3 current?

We thank the reviewer's thoughtful comments regarding the mutagenesis analysis and the possible mechanisms underlying zinc regulation of Slo3. Regarding the suggestion to perform double or combination mutants, we agree that such experiments would provide valuable mechanistic insight. However, due to limited resources, we were not able to perform these additional experiments within the scope of this study. Our current results show that mutations at E169 and E205 partially abolish zinc inhibition, which suggests that the inhibitory mechanism is not mediated through a single residue and is likely more complex.

Alternative mechanisms that may contribute to zinc modulation of Slo3 include indirect effects through modulation of nearby charged residues, structural rearrangements influenced by zinc binding, or the presence of multiple zinc binding sites within Slo3 channel other than the sites discovered through this study. At present, these mechanisms remain speculative and further studies will be required to clarify their contributions. This study provides the foundational basis for understanding how zinc inhibits the Slo3 channel and serves as an important starting point for defining the molecular mechanism in more detail.

We already acknowledged in the Discussion section that the precise molecular basis of zinc inhibition remains unknown and that future work involving more extensive mutational and structural analyses will be essential to fully resolve this issue.

We also added the discussion section as follows:

"It is worth noting that the incomplete loss of zinc sensitivity in these mutants suggests that additional mechanisms may participate in zinc modulation of Slo3. These may include modulation of nearby charged residues, structural rearrangements influenced by zinc binding, or the presence of multiple zinc binding sites. Comparisons with Slo2.2 (J. Zhang et al., 2023), KCNQ4 (Gao et al., 2017), and voltage-gated calcium channels (Sun et al., 2007) further support the possibility of diverse molecular determinants for zinc inhibition. Our VCF, mutagenesis, and simulation data together indicate that zinc influences voltage sensor

movement in mSlo3, which may suggest a distinct inhibitory mechanism that warrants further investigation.”

While elucidating the mechanism of Slo3 is interesting, there is substantial literature indicating how zinc regulates channel functions at a molecular level. Given this, the manuscript should provide a deeper understanding by clearly elucidating the molecular mechanism of the regulation of Slo3 current by zinc.

Thank you for highlighting a very important point that requires deeper discussion and explanation regarding how zinc regulates Slo3 current at the molecular level. As reported, Slo3 is gated by membrane depolarization and, at the same time, this channel is also gated by intracellular pH, particularly alkalinization (Leonetti et al., 2012; Schreiber et al., 1998; X. Zhang et al., 2006). This makes the gating mechanism of this channel complex. The molecular mechanism underlying pH regulation of the Slo3 channel remains unknown (M. D. Lyon et al., 2023). We tested different pH conditions and membrane voltage to elucidate the effect of zinc on the Slo3 channel. Our data suggests that zinc inhibition in mSlo3 channels is dependent on pH (Fig. 2A-E), voltage (Fig. 2G-H; Fig. 2—figure supplement 1A, B) and exhibits a long-lasting inhibitory effect (Fig. 2I, K).

However, as much as we are aware that these data alone cannot explain the molecular mechanisms of zinc’s effect on Slo3 current, our mutagenesis experiments also did not provide a straightforward answer. The single amino acid mutations examined in this study, which contain clustered negative residues, did not significantly alter zinc-mediated current reduction compared to the wild type. As the reviewer pointed out, mutating one single amino acid may not be sufficient to fully identify other contributing residues within the predicted mSlo3 zinc-binding site. Therefore, more extensive mutagenesis studies will be required to fully elucidate the molecular mechanism of zinc inhibition in mSlo3, which could not be fully understood in this study.

On the other hand, when we analyzed the percentage of current recovery of all the mutants, E169A and E205A showed significant current recovery upon the wash-out by pH 8.0 alone. Consistent with MD simulations, our electrophysiological recordings demonstrated that the long-lasting inhibitory effect of zinc was partly abolished by these mutations. Thus, our findings highlight the contribution of E169A, located at the lower end of S3 domain and E205A, located at the lower region of S4 domain, to zinc-mediated inhibition of mSlo3 current.

Additionally, since the molecular mechanism of pH regulation on Slo3 channel remains unknown, the molecular basis of its dual gating has yet to be elucidated, making it difficult to draw a single definitive conclusion from our current research data on how zinc inhibits mSlo3 current. Nevertheless, this study provides the foundation for understanding possible mechanisms of zinc inhibition. Our VCF data suggest that zinc influences the movement of VSD of mSlo3, and together with our mutagenesis and MD simulations results, these findings represent an important first step toward elucidating the molecular mechanism of zinc inhibition of the mSlo3 current.

Intracellular zinc exerts inhibitory effect on mSlo3, similar to what has been reported for Slo2.2 channels (J. Zhang et al., 2023), high- and low-voltage activated calcium channel families (Sun et al., 2007) and KCNQ4 channels (Gao et al., 2017). These studies identified different regions, amino acids, and possible mechanisms of zinc inhibition among these ion channels. For instance, in Slo2.2 channels, which belong to the same Slo family as Slo3, the zinc-binding site was identified in the RCK2 domain, where cysteine and histidine residues form a canonical zinc binding motif (J. Zhang et al., 2023). In KCNQ4 channels, zinc inhibits the channel activity in a non-canonical manner that depends on its physiological activator, the membrane lipid PI(4,5)P₂ (Gao et al., 2017). Although zinc exerts the inhibitory effects on those various voltage-gated potassium and calcium channels, the mechanisms differ. Our

data suggests another distinct mechanism of zinc inhibition in the mSlo3 channel with the identified sites located in the VSD, where zinc influences the voltage-sensor motion, and consequently affects the complex gating of Slo3.

We revised the discussion section as follows, which is also related to the previous comment:

“It is worth noting that the incomplete loss of zinc sensitivity in these mutants suggests that additional mechanisms may participate in zinc modulation of Slo3. These may include modulation of nearby charged residues, structural rearrangements influenced by zinc binding, or the presence of multiple zinc binding sites. Comparisons with Slo2.2 (J. Zhang et al., 2023), KCNQ4 (Gao et al., 2017), and voltage-gated calcium channels (Sun et al., 2007) further support the possibility of diverse molecular determinants for zinc inhibition. Our VCF, mutagenesis, and simulation data together indicate that zinc influences voltage sensor movement in mSlo3, which may suggest a distinct inhibitory mechanism that warrants further investigation.”

The manuscript includes no experimental data on the mechanism of intracellular zinc export during sperm capacitation, despite being crucial for the regulation of sperm function.

We thank the reviewers for the valuable comment in this regard. We agree that mechanism of intracellular zinc export during capacitation is crucial for the regulation of sperm function, and it would be an important finding if we could provide the experimental data on this. However, there are significant technical difficulties in performing such experiments. Two protein families facilitate the transport of zinc across cellular and intracellular membranes in opposite directions: ZnT and ZIP. ZIP12 has been reported to be highly expressed in mouse testis (Zhu et al., 2022), as well as ZnT-1 (Elgazar et al., 2005). To date, there are no known inhibitors for zinc transporters, and there is also no suitable antibodies available for these transporters, which makes it difficult to design experiments to examine the intracellular zinc transport during sperm capacitation. Apart from the two reported zinc transporters, the functional significance of other ZnTs and ZIPs, particularly those related to capacitation, remains largely unclear, leaving the mechanisms of zinc transport in sperm during capacitation poorly understood. Moreover, homozygous Znt-1 knockout mice exhibit a lethal phenotype (Andrews et al., 2004).

Reviewer #2 (Public review):

Summary:

In this paper, Andriani and colleagues are examining the potential role of Zn flux in sperm and its effect on Slo3 channels. This is an interesting question that is likely critical to how sperm function properly and Slo3 channels are a possible candidate for a downstream molecule that is impacted by Zn. In this paper, the authors use Zn imaging, sperm motility assays, and electrophysiology to show that Zn flux impacts sperm function. They then go on to look at the impact Zn has on Slo3 current and propose a binding site based on MD simulations. While the ideas are interesting, the experiments are not well described in many places making understanding the results very difficult. In addition, critical controls are missing throughout the paper.

Strengths:

The question of how Zn flux impacts membrane potential and sperm motility is an important one. Moreover, Slo3 presents an interesting candidate or the target of Zn regulation. The combination of methods used here also has the potential to uncover mechanisms of Zn regulation of Slo3.

Weaknesses:

Much of the paper lacks experimental description which makes interpretation quite difficult, or a detailed discussion is missing. Examples include:

(1) Figure 1, particularly the Zn imaging, is not sufficiently described. How is the fluorescence intensity measured? A representative ROI? The whole tail and head? Are the sperm immobile? If not, there is evidence that motion artifacts can significantly distort these sorts of measures from Calcium measurements in Cilia. Were there controls done? Is the small amount of Zn seen in the tail above the background?

We sincerely thank the reviewer for pointing out important details that we should provide in this study in order to make it well understood. We would like to answer and respond to the points raised by reviewer as follows:

Fluorescence intensity is measured by the signal taken from the whole head and the proximal part of tail in sperm. We have included this in the materials and methods.

Materials and Methods

“Fluorescence intensity is measured by the signal taken from the whole head and the proximal part of tail in sperm.”

Yes sperm is immobile during zinc imaging.

We added the control data of zinc imaging without capacitation medium and incorporated the data into the graph in Figure 1B. For the control in non-capacitation medium, we use HS medium as newly explained in the methods, results, related figure (Figure 1B), and figure legends.

Yes the small amount of Zn seen in the tail above the background. As shown in Fig. 1A we confirmed that the signal intensity at the proximal region of the tail was higher than the background. Therefore, the data for this region were calculated after background subtraction.

(2) The second half of Figure 1 is also not well described. What is the extracellular solution in the recordings? When you apply the Zn ionophore, do you expect influx or efflux? I assume efflux is based on the conclusions but this should be discussed explicitly.

The extracellular solution in the recordings for Figure 1 is HS solution (HEPES-buffered saline solution), a standard non-capacitation medium. We will include this information in the materials methods.

Materials and methods

“HS-based solution was used as the extracellular solution.”

We assume that intracellular zinc levels increase upon application of zinc ionophore. Previous work has reported that sperm contain approximately 35.7 ng/10⁶ cells in the head and flagellum (Henkel et al., 1999). When zinc pyrithione is applied, it facilitates the influx of Zn²⁺ from the surrounding medium into the cell, thereby increasing intracellular zinc concentration. Zinc pyrithione functions both as a zinc source and as a transport facilitator, allowing Zn²⁺ to cross the otherwise impermeable lipid membrane without compromising membrane integrity.

(3) Figure 2H labels the Y axis, "normalized current". Normalized to what? Why do neither of the curves end at 1? A better description of what this figure represents is needed.

Normalization for figure 2H was performed by dividing the absolute current of mSlo3 at pH 8.0 of each voltage by the absolute current at the pre-determined highest voltage that still produced a stable mSlo3 current (i.e., good patch, good clamp). In this analysis, +140 mV was

chosen as the highest voltage for normalization, since in few cells the patch was lost at +160mV and +180mV. Similar to the control condition, the absolute current of mSlo3 in the presence of 100 μ M zinc was normalized to the absolute current of the control at +140 mV. This information has been included in the figure legends and the Materials Methods section of the revised manuscript.

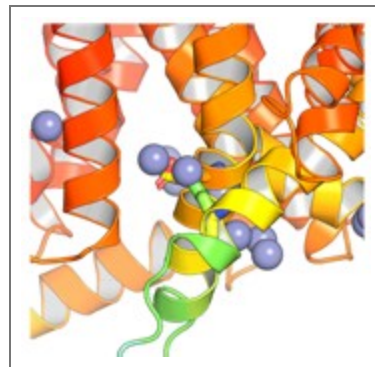
Materials Methods section:

Figure legends for figure 2H has been updated.

(4) The alpha fold simulations are not well described. How many Zn binding sites were found? Are all of the histidine mutations in Figure 4 Supplement 1 the ones that were found?

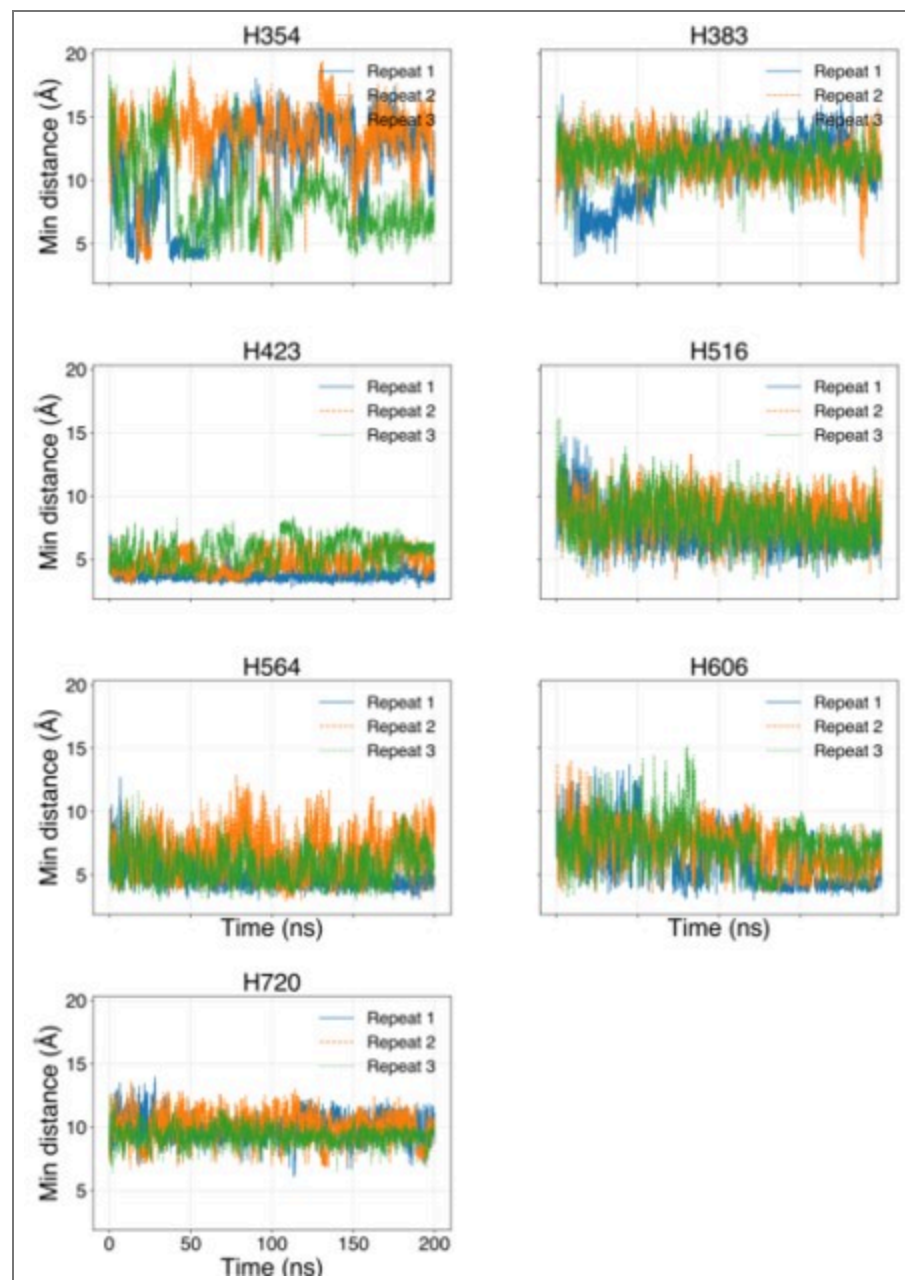
We thank the reviewer for the question. In our AlphaFold3 input, we only input the transmembrane region of the protein. From there, we found four sites located as follows:

Given that we are only interested in the intracellular side of the membrane, we are only interested in the site with the highest pLDDT value (confidence values). On the IC side, there are only two sites, where the other sites are located near the pore domain. The site is near E310 and K319.



Author response image 1. AlphaFold3 prediction of the Zn binding site on IC side of Slo3

The histidines in Fig. 4—figure supplement 1 are all histidines that are not in the transmembrane region. These residues were not included in the initial inputs for AlphaFold3. However, we conducted MD simulations including these residues and we were able to show that a few of these residues are in contact with Zn. We have now plotted the minimum distance between each of these residues and Zn in the flooding simulations.



Author response image 2. MD simulations of histidines residues located in IC of Slo3

Minimum distances between histidines in Fig. 4—figure supplement 1 and Zn^{2+} from the flooding simulations. Different colors indicate different repeats.

(5) *There is no discussion of physiological intracellular Zn concentration. How much Zn is inside the sperm? How much if likely Free vs buffered? Is 100uM a reasonable physiological concentration?*

We estimated the intracellular zinc concentration in sperm based on human sperm data, which report a zinc concentration of approximately $35.7 \text{ ng}/10^6 \text{ cells}$ in the head and flagellum (Henkel et al., 1999). Considering the volume of a typical human sperm is about $15 \text{ } \mu\text{m}^3$ (Laufer et al., 1977), this translates to an estimated intracellular zinc concentration of approximately 400 mM, although the concentration of free zinc must be much lower than this level. Although exact intracellular zinc concentrations in mouse sperm are not well-

documented, this estimate supports the observation of elevated zinc in non-capacitated sperm.

There are a number of areas where the interpretation is not well supported by the data including:

(6) You say in the Figure 4 supplement, that "we did not observe any significant decrease in the percentage of current inhibition." But that is a pretty misleading statement. There are large changes (increases) in the amount of zinc inhibition. These might be allosteric changes but I don't think you can safely eliminate these as relevant Zn binding sites. Also, some of these mutations appear to allow at least some unbinding of Zn.

In our MD simulations, H720 is not at the zinc binding site and therefore, mutation to arginine would indeed eliminate its binding. We are showing this in the minimum distance analysis between Zn and H720 and show that they are further than 4 Å from each others (n=3), as shown in author response image 2.

Chimera of Slo3/Slo1 RCK2 also showed large increases in the amount of zinc inhibition, and this might serve as a potential binding site. We agree that the statement: "we did not observe any significant decrease in the percentage of current inhibition." is misleading, therefore we revised our interpretation and statement into:

We revised the result section as follows:

"However, the percentage of current inhibition varied across the mutated constructs, showing either increases or no appreciable change (Fig. 4—figure supplement 1B, C)."

(7) Following up on the above point, it seems unfair to conclude that the D162S, E169A, and E205 mutants are part of the inhibitory binding site for Zn when the mutation has no effect on inhibition and only an effect on the washout. The mutations on the intracellular side also had an impact on the washout so it seems equally likely that they are the critical residues based on your data.

We thank the reviewer for this important point. We agree that the absence of a strong reduction in the initial zinc inhibition makes it challenging to assign any single residue as a definitive zinc binding site. However, our interpretation is based not only on the electrophysiological data but also on the MD simulations, which consistently identified E169 and E205 as residues that frequently interact with zinc and stabilize zinc occupancy within the VSD region. Although the mutations did not markedly reduce the peak level of zinc inhibition, both E169A and E205A significantly altered the long-lasting inhibitory component during washout, which is consistent with the MD-predicted interactions. In contrast, the intracellular mutations affected washout but were not supported by MD simulations as potential zinc interaction sites. Taken together, these combined datasets support the idea that E169 and E205 contribute to zinc modulation of Slo3 in the VSD, even though additional residues or mechanisms are likely involved.

(8) Nowhere in the paper do you make the specific link between Zn flux and membrane hyperpolarization via Slo3. You show that Zn flux changes the ability of the sperm to hyperpolarize and you show that Slo3 is inhibited by Zn but the connection between the two is not demonstrated. There appears to be a specific Slo3 blocker. If you use this in sperm, do you no longer see the Zn effect?

Thank you for pointing out the need for clarifying this point. It is already known that sperm capacitation is well associated with the increase of intracellular pH (Vredenburg-Wilberg & Parrish, 1995; Y. Zeng et al., 1996), the hyperpolarization of the membrane (Arnoult et al., 1999; Y. Zeng et al., 1995) and the elevation of intracellular Ca^{2+} concentration level (Breitbart, 2002; Publicover et al., 2007) through diverse ion channel activities. To explore

whether these pathways are influenced by intracellular zinc, we used patch-clamp techniques to measure the membrane potential (V_m) as shown in Fig. 1D-K. It has been reported that under the whole-cell current clamp of mouse epididymal spermatozoa, resting membrane potential is hyperpolarized after intracellular alkalinization (Navarro et al., 2007). We mentioned this in line 100-108 in the manuscript.

Next, our findings from the experiments using mouse spermatozoa suggest that intracellular zinc inhibits a key process in sperm capacitation, specifically the alkalinization-induced hyperpolarization. Previous studies have identified the pH- and voltage-dependent potassium channel Slo3 is responsible for the principal K^+ current ($I_{K_{Sper}}$) in mouse spermatozoa (Navarro et al., 2007; Santi et al., 2010; Schreiber et al., 1998; X. H. Zeng et al., 2011). During capacitation, the rise in pH_i leads to the activation of Slo3 channels, resulting in membrane hyperpolarization (Santi et al., 2010). Given this context, we next investigated whether intracellular zinc acts directly on the Slo3 channel and found that zinc inhibits mSlo3 current. We explained this rationale of the experiment in line 143-150.

We add following sentence to add more clarity to the text:

“During capacitation, the rise in pH_i leads to the activation of Slo3 channels, resulting in membrane hyperpolarization (Santi et al., 2010).”

Therefore, the text was modified into:

“Our findings suggest that intracellular zinc inhibits a key process in sperm capacitation, specifically the alkalinization-induced hyperpolarization. Previous studies have identified the pH- and voltage-dependent potassium channel Slo3 is responsible for the principal K^+ current ($I_{K_{Sper}}$) in mouse spermatozoa (Navarro et al., 2007; Santi et al., 2010; Schreiber et al., 1998; X. H. Zeng et al., 2011). During capacitation, the rise in pH_i leads to the activation of Slo3 channels, resulting in membrane hyperpolarization (Santi et al., 2010). Given this context, we next investigated whether intracellular zinc acts directly on the Slo3 channel.”

Regarding the specific inhibitor, as has been pointed out by the reviewer that a new Slo3 inhibitor, VU0546110, exhibited more than 40-fold selective for human Slo3 over Slo1 (M. Lyon et al., 2023). However, the effect of VU0546110 in mSlo3 has not been tested yet. Both mouse and human Slo3 exhibit similar responses to certain inhibitors, but mouse and human Slo3 is also differ in their responses to several other inhibitors (M. D. Lyon et al., 2023), making it uncertain if this VU0546110 will work on mSlo3.

(9) In the second half of Figure 1, the authors suggest that there is “no hyperpolarization in 100 μM Zn. That is not really true. It is reduced but not absent.

We modified the wording of “no hyperpolarization in 100 μM Zn” to “alkalinization-induced hyperpolarization was reduced in the 100 μM $ZnCl_2$ group.”

“In contrast, alkalinization-induced hyperpolarization was reduced in the 100 μM $ZnCl_2$ group”

(10) The claim that *Lrcc52* with Slo3 shows a higher current inhibition at pH 7.5 than pH 8 is not well supported because there are only 3 replicates in the 7.5 case. In addition, the claim is made in the test that 100 μM $ZnCl_2$ “already inhibited mSlo3+*Lrcc52* at pH 7.5”, contrasted with mSlo3 alone, is not tested statistically.

Thank you for the valuable comment. Although Fig. 3F shows a statistical difference, we agree that having only three replicates at pH 7.5 may somewhat weaken the conclusion. Following this suggestion, we have revised the sentence as follows:

“Alkalinization appeared to increase the percentage of current inhibition by 100 μM $ZnCl_2$.”

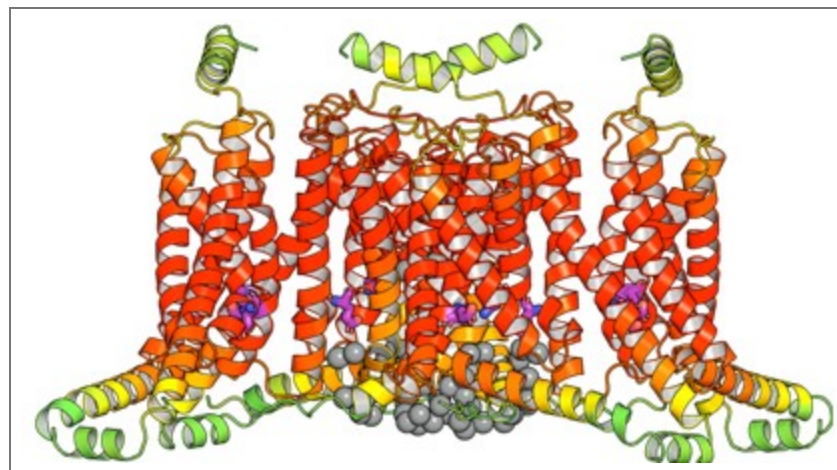
We provided statistical analysis to compare pH 7.5 between mSlo3 alone and mSlo3+Lrrc52 in the Figure 3—figure supplement 1D:

The statistical analysis showed that 100 μM zinc significantly inhibited the mSlo3 + Lrrc52 current at pH 7.5 compared to the mSlo3 current alone. We have incorporated the necessary changes into the revised manuscript and updated the figure legends accordingly.

In a number of places, better controls are needed.

(11) How specific is this effect for Zn? Mg^{2+} , for instance, is also a divalent cation that is in the hundreds of μM range inside the cell. Does it exert the same effect? Each ion certainly has unique preferred coordination geometries, does your predicted binding with MD show what you might expect for tetrahedral coordination with Zn? Did you test other divalent cations functionally or in silicon?

To answer this question, we have tested this by building another AlphaFold3 model, with Mg^{2+} instead of Zn^{2+} . We did not opt for the all-atoms MD simulations due to the cost of the simulation. Here, the model shows that Mg are all clustered at the pore domain and does not reside anywhere near the Zn^{2+} site from both MD simulations and the AF3 model.



Author response image 3. AlphaFold3 model of Slo3 channel with Mg^{2+}

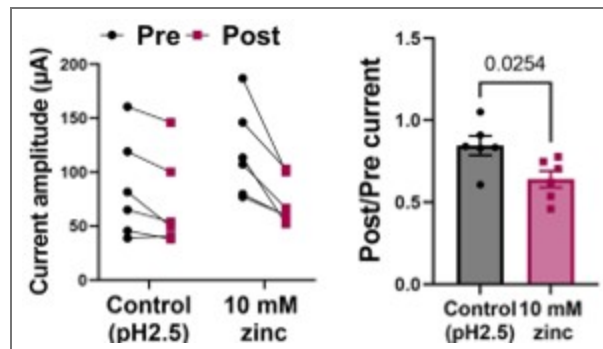
The Slo3 AlphaFold model from residue M1 to L330. The colour gradient reflects the pLDDT score range from 1.73 to 95.69. Purple sticks highlighted E169, N171 and E205. In this study, we did not examine other divalent cations in our electrophysiological recordings. Exploring their effects will be an important direction for future research.

(12) For the VCF experiments, a significantly higher concentration of Zn was used (10mM). What is the reason for this? There is no discussion of how much a "puff" is. Assuming you are using the RNA injector it is probably on the order of 50nL or less. Assuming the volume of an oocyte is 1uL that would argue that the final concentration is 500uM or higher. But this is also complicated by potential local effects of high Zn at the injection site, artifacts of injecting that much metal, and the fact that a great deal of the Zn will likely be bound to other things inside the cell. Better controls are needed for this experiment.

As pointed out by the reviewer, the volume of the oocytes is estimated to be approximately 1 μL . We performed manual injections using glass needle typically used for RNA injection. However, because the injections were done manually during real-time VCF recording (as illustrated in the experimental scheme), the exact volume of the solution injected into each

oocyte could not be precisely controlled. We estimated that each drop to be approximately 50 nL, resulting in a final concentration around 500 μ M, as described by the reviewer.

The rationale for using relatively high concentration was to ensure that the zinc concentration inside the oocyte reached an effective level, since manual injection may sometimes deliver less than 50 nL of solution. In some cases, injections failed entirely due to the technical difficulty of the method. Because VCF recordings are already technically difficult, we aimed to ensure that zinc injection was successful in oocytes that exhibited robust fluorescence signal by injecting an excess amount of zinc that would not disrupt normal oocyte conditions. For example, 10 mM zinc was prepared in an acidic solution (pH 2.5). We verified that this acidic condition did not affect mSlo3 current by performing control injections with the acidic solution alone, since the mSlo3 current is not activated under acidic pH conditions



Author response image 4. VCF control experiments: vehicle injection.

Reviewer #3 (Public review):

Summary:

The study titled "Zinc is a Key Regulator of the Sperm-Specific K⁺ Channel (Slo3) Function" aims to investigate the role of intracellular zinc in sperm capacitation and its regulation of the sperm-specific Slo3 potassium channel. Capacitation is a crucial physiological process that enables sperm to fertilize an egg, and membrane hyperpolarization through Slo3 activation is a well-established event in this process. The authors propose that intracellular zinc dynamically decreases during capacitation and inhibits Slo3-mediated K⁺ currents, thereby playing a regulatory role in sperm function.

Strengths:

(1) Novel Contribution to Sperm Physiology.

The study provides new insights into how zinc dynamics contribute to sperm capacitation, specifically through its direct inhibition of Slo3 activity.

Previous research has focused primarily on extracellular zinc's effect on sperm function; this work expands the discussion to intracellular zinc regulation, an area with limited prior investigation.

(2) Strong Electrophysiological Evidence.

*The study employs inside-out patch-clamp recordings in *Xenopus* oocytes to demonstrate zinc's direct inhibition of Slo3 currents. The observed slow dissociation of zinc from Slo3 suggests a long-lasting regulatory effect, adding to the understanding of ion channel modulation in sperm cells.*

(3) Molecular Mechanistic Insights

Using Molecular Dynamics (MD) simulations and mutagenesis, the authors identify potential zinc-binding sites within Slo3's voltage-sensing domain (VSD), particularly E169 and E205. These computational predictions are supported by electrophysiological recordings, strengthening the argument that zinc directly binds and inhibits Slo3.

(4) Physiological Relevance and Functional Implications

The study suggests that zinc inhibition of Slo3 could contribute to sperm motility regulation during capacitation.

The authors provide sperm motility assays as supporting evidence, showing that zinc chelation affects motility only after capacitation has begun, suggesting a dynamic role of intracellular zinc in the capacitation process.

Weaknesses:

While the study presents compelling electrophysiological data and molecular insights, there are several critical gaps that must be addressed before fully supporting the physiological relevance of the findings.

(1) The authors should measure the effects in sperm cells using the patch-clamp technique to directly record Slo3 currents. By normalizing Slo3 currents to cell capacitance at different intracellular zinc concentrations, the authors can quantitatively assess the extent of Slo3 inhibition by zinc and strengthen the physiological relevance of their findings.

We thank the reviewer for the valuable comments to strengthen the physiological relevance of our findings. We provided additional data of Slo3 currents measured using perforated patch-clamp recording in sperm cells in experiments with zinc pyrithione (ZnPy) before and after the addition of 10 mM NH₄Cl. Control experiments were conducted in the absence of ZnPy, in which Slo3 current were recorded before and after the application of 10 mM NH₄Cl. These data have been integrated into Figure 1L-N and Figure 1—figure supplement 1A, B.

It is worth noting that Slo3 current in this recording might contain other endogenous current, as no specific blocker was used. Nonetheless, the data showed that the Slo3 current in sperm tends to be inhibited by zinc, as shown by the plot of absolute Slo3 current after the addition of 10 mM NH₄Cl in the absence of ZnPy (control) and in the presence of 100 μM ZnPy. There was a decrease in the fold change calculated from the absolute current before and after the addition of 10 mM NH₄Cl of ZnPy treated group compared to the control group.

We also provided data with the cell capacitance as suggested; however, cell capacitance obtained from the sperm recordings showed the capacitance throughout the head and midpiece of spermatozoa. On the other hand, Slo3 channels are not expressed in the entire spermatozoa, therefore the cell capacitance acquired from these recordings does not accurately reflect the area where the Slo3 channels are localized. Although we included normalization of Slo3 currents to cell capacitance before and after ZnPy application, this normalization should be interpreted with caution for the reasons mentioned above. The corresponding figure has been included in the supplementary data Figure 1—figure supplement 1A, B.

We added sentences to the result section as follows:

“We also measured Slo3 current using perforated patch-clamp recordings in spermatozoa treated with ZnPy, before and after the addition of NH₄ Cl. Control experiments were conducted in the absence of ZnPy, in which Slo3 current were recorded before and after the

application of 10 mM NH_4Cl (Fig. 1L-N; Fig. 1—figure supplement 2A, B). Slo3 current in sperm tended to be inhibited by zinc, as shown by the plot of absolute Slo3 current after the addition of 10 mM NH_4Cl in the absence of ZnPy (control) and in the presence of 100 μM ZnPy (Fig. 1L, M). There was a decrease in the fold change calculated from the absolute current before and after the addition of 10 mM NH_4Cl of ZnPy treated group compared to the control group (Fig. 1N). Taken together, these results confirmed that intracellular zinc indeed inhibits alkalization-induced hyperpolarization in mouse sperm.”

(2) Lack of Controls in Non-Capacitated Sperm

The claim that zinc is exported from sperm during capacitation needs stronger experimental validation.

The authors did not include a control group of non-capacitated sperm in key fluorescence imaging experiments, making it difficult to confirm that the observed zinc decrease is capacitation-specific rather than a general zinc redistribution process.

To strengthen this conclusion, experiments should be performed in non-capacitating conditions to determine whether intracellular zinc levels remain unchanged.

We added the control group of non-capacitated sperm in key fluorescence imaging experiments, as integrated in Figure 1B.

The following changes in the Results and Figure Legend sections are revised and added:

“We observed that there was a gradual and significant decrease in fluorescence intensity in both regions (Fig. 1B), particularly prominent in the flagellum (Fig. 1C). This decline suggests the active release of intracellular zinc from sperm flagellum occurs during capacitation. In contrast, the fluorescence intensity of the control group of non-capacitated sperm remained unchanged (Fig. 1B).”

Figure Legend 1B was modified accordingly.

(3) Unclear Role of Zinc in Physiological Capacitation

The study clearly demonstrates zinc inhibition of Slo3 but does not sufficiently establish how this affects capacitation at a functional level.

Additional motility and capacitation markers should be analyzed to confirm that zinc influences sperm behavior beyond Slo3 inhibition.

We thank the reviewer for this valuable comment. We fully agree that zinc can influence sperm physiology through multiple mechanisms and that its overall effects on capacitation are complex. However, the main goal of our study is to investigate the mechanism and to determine whether intracellular Zn^{2+} directly inhibits Slo3. Our results from both the heterologous expression system and the sperm membrane potential recordings consistently support this conclusion.

For these reasons, we believe that adding such assays would not clarify the role of Slo3 in capacitation but rather risk confounding interpretation. Instead, we have expanded the Discussion to explicitly acknowledge these limitations and to emphasize that future studies combining genetic or pharmacological modulation of Slo3 with comprehensive capacitation analyses will be required to fully define its physiological impact.

We added sentences to the discussion section in the revised manuscript as follows:

“Although these results support a mechanistic link between zinc and Slo3 activity, future studies that combine genetic or pharmacological modulation of Slo3 with comprehensive capacitation analyses will be required to define its physiological impact in more detail.

Within this context, this study highlights the potential importance of intracellular zinc in the regulation of sperm capacitation.”

(4) Insufficient Data on Zinc-Slo3 Specificity

The authors should consider using quinidine, a known washable Slo3 inhibitor, to confirm that zinc acts specifically on Slo3 channels rather than other endogenous ion channels.

The study would benefit from including washout controls in the inside-out patch-clamp recordings, as seen in Figure 3-Supplement 1, to confirm that zinc inhibition is reversible or long-lasting.

We thank the reviewer for raising the point regarding the need to confirm that the current observed in our recordings indeed represents Slo3 current by using a specific blocker such as quinidine, as there is a possibility that endogenous currents might also be present and that zinc could act on those endogenous currents. Performing experiments with quinidine would indeed be crucial to demonstrate the specificity of Slo3 current in our patch-clamp recordings.

However, in our current experimental protocol, we apply ramp pulses multiple times and require a long series of recordings within a single session in one patch as described in the materials and methods as well as Figure 2I, Figure 4—figure supplement 1C, Figure 5B (pH 8.0 → 100 μ M zinc → pH 8.0, to observe the washout effect). Incorporating quinidine into this sequence would make the protocol even longer (pH 8.0 → quinidine → washout → pH 8.0 → 100 μ M zinc), which increases the likelihood of patch loss before completing the full set.

Furthermore, we have ensured that the recorded current corresponds to Slo3 by using appropriate experimental conditions, specifically the suitable voltage range for activation, a high intracellular pH (pH 8.0), and high-potassium solutions in our recordings.

(5) Missing Discussion of Zinc's Role in CatSper Regulation

The study focuses solely on Slo3 but does not mention CatSper, the principal Ca^{2+} channel essential for sperm capacitation.

Zinc has been reported to inhibit CatSper activity, which could significantly impact sperm function.

The discussion should address whether zinc's effect on Slo3 represents a broader regulatory mechanism influencing multiple ion channels during capacitation.

Thank you for the comment. To the best of our knowledge, there have been no reports showing that CatSper activity is directly regulated by zinc ions.

Furthermore, in our patch-clamp recordings with NH_4Cl and ZnPy, we observed that the normal CatSper current increased even in the presence of ZnPy, which makes it challenging to conclude whether zinc directly affects CatSper channel activity.

We added sentences to the discussion section in the revised manuscript as follows:

“In addition to that, to date, there are only few reports on the effect of zinc on other sperm ion channels, and none have been reported in mouse sperm. One important study was reported by (Jeschke et al., 2021), in which seminal zinc was found to inhibit prostaglandin-induced activation of CatSper, a sperm-specific Ca^{2+} channel, in human sperm. The complex opposing action of seminal zinc and prostaglandins on CatSper may help preventing premature activation of CatSper in the ejaculate and act as a dilution sensor, although this

study does not provide direct evidence for zinc acting directly on CatSper (Jeschke et al., 2021)."

Final Assessment

This work presents important findings on zinc regulation of Slo3 channels, supported by strong electrophysiological and molecular analyses. However, the physiological relevance of these findings remains unclear due to missing controls, and needs additional functional assays. Addressing these issues would significantly enhance the manuscript's scientific rigor and impact.

Recommendations for the authors:

Reviewer #2 (Recommendations for the authors):

Most of the specific comments and suggestions are in the public review. Minor additional comments primarily focused on presentation and textual errors are here.

(1) There is something strange happening in Figure 6D in the -100ish range. I think it's likely related to the reversal potential of K⁺.

Thank you for pointing it out. Yes in figure 6D there was strange plot in the range of -100 mV. As the reviewer has pointed out we also think that it is related to the reversal potential of potassium ions.

(2) There are a number of errors in the text that make following it difficult. For instance, multiple times the authors say "In consistent" (line 120 as an example) when I think they mean consistent with.

We changed the "in consistent" with "consistent with" throughout the revised manuscript.

Reviewer #3 (Recommendations for the authors):

The authors provide well-described experiments, particularly those examining the effects of intracellular zinc on Slo3 channels using inside-out patch-clamp recordings. However, some experimental designs intended to assess the physiological relevance of these findings during capacitation require additional controls and data before the authors' claims can be fully supported.

Comments

Major Concerns & Suggested Improvements

Line 65: "In the present study, we find that intracellular zinc is exported during capacitation, indicating that zinc dynamics in spermatozoa play an important role in fertilization."

This claim requires additional experimental data to be fully supported.

Thank you for pointing it out. We have provided data for control experiments of zinc imaging in non-capacitated conditions in Figure 1B.

Line 79: "Intracellular zinc is exported from sperm during capacitation."

The authors should include controls in non-capacitated conditions to determine whether zinc export is specific to capacitation or a general process in sperm cells.

Again, we have provided data for control experiments of zinc imaging in non-capacitated conditions in Figure 1B.

Figures - General Comment:

In all figures, please replace SEM (Standard Error of the Mean) with Standard Deviation (SD) for consistency and a more accurate representation of variability.

SEM (Standard Error of the Mean) has been replaced with SD (Standard Deviation) in all figures (main figures and supplements) as well as in numerical description accordingly.

Figure 1

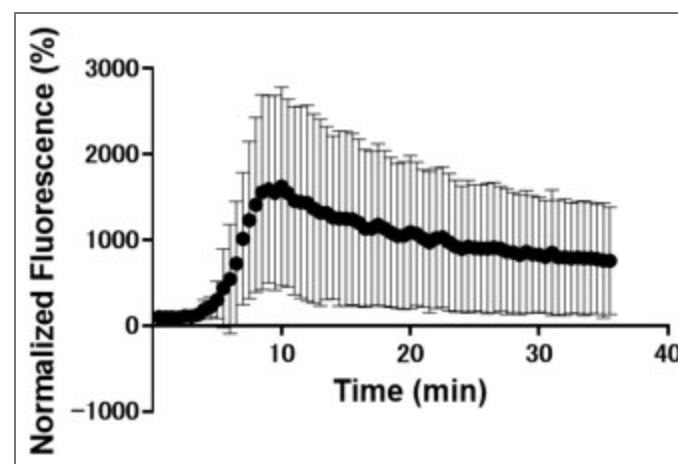
Panel B:

Include a non-capacitating media control to confirm that the observed decrease in zinc-sensitive dye fluorescence is not due to artifact/photobleaching.

We have provided data for control experiments of zinc imaging in non-capacitated conditions in Figure 1B.

Perform an experiment with capacitating media supplemented with a higher concentration of zinc. If intracellular zinc export is a real effect, added extracellular zinc should prevent or reduce this phenomenon.

We appreciate the reviewer's suggestion; however, we believe that supplementing the medium with high concentrations of zinc is unsuitable for validating the export phenomenon due to confounding physiological factors. Our preliminary tests demonstrated that increasing extracellular zinc triggers a drastic increase in intracellular zinc as well (Author response image 5). Furthermore, the high concentration of BSA in the capacitation medium acts as a potent zinc buffer, precluding precise control over free Zn^{2+} levels. Therefore, the inherent difficulty in maintaining defined extracellular and intracellular Zn^{2+} gradients makes the interpretation of such data highly problematic. Future studies will focus on identifying the specific zinc transporters involved and characterizing their molecular mechanisms.



Author response image 5. Zinc addition

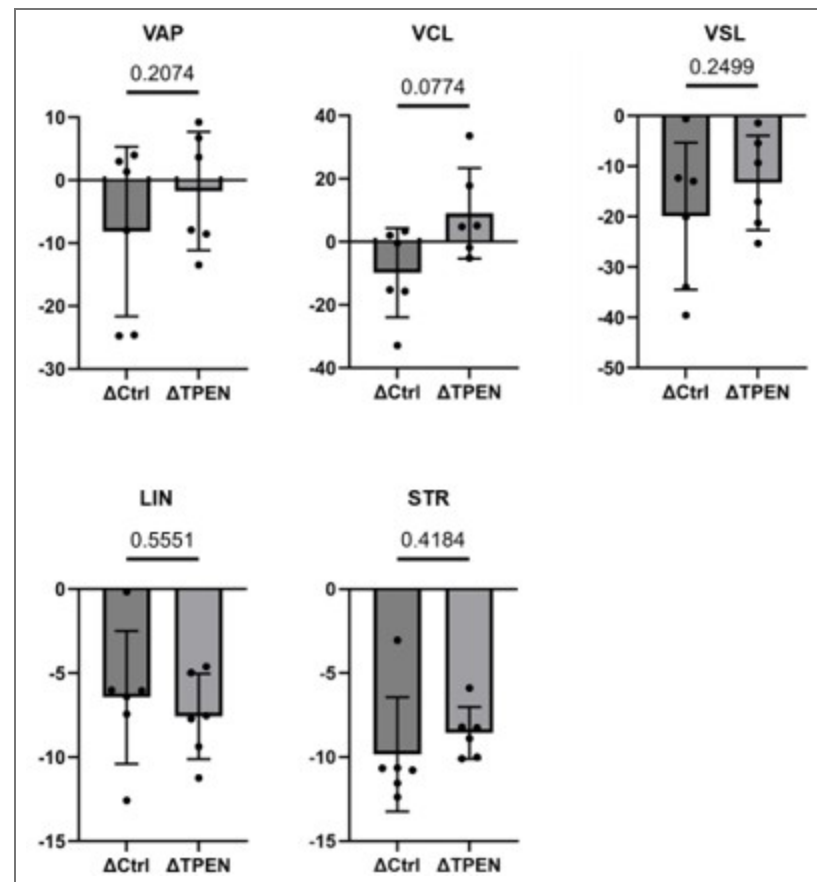
Clarify whether the "n" value represents different cells or multiple recordings from the same cell.

n value represents different cells.

Supplemental Figure 1:

Incorporate Δ (delta) comparison between 10 min and 2 hours under control conditions and in the presence of TPEN.

Here we provide data:



Author response image 6. Δ comparison between control and TPEN

Provide statistical analysis for these comparisons to make the effects of capacitation clearer.

We did the calculation and statistical analysis, however there was no statistical difference, as shown in the author response figure 6 due to high variability of individual data.

Figure 2

Panel C:

Incorporate inhibition at pH 7.4 and 6.0 for direct comparison.

Recording inhibition effect of zinc at pH 6.0 is not possible because there would be no current to begin with, as mSlo3 is gated by both voltage and alkaline pH.

Panel D:

Include a washout control, similar to what is shown in Panel A.

We included a washout control trace to Figure 2D.

Panel E:

Provide a longer reference trace in the absence of zinc to clearly visualize the control condition. The current reference segment is too short to properly assess baseline activity.

Although we do not have a longer reference trace in the absence of zinc for Figure 2E, we instead show the trace recorded under the application of 0.1 μ M zinc in Figure 2—figure supplement 1A to illustrate the current behavior.

Panels G-H:

Include inside-out patch-clamp traces and quantification of zinc washout effects.

Inside out patch traces are shown in Figure 2G as we applied step-pulses protocol. The zinc washout effect could not be quantified because the patch was usually lost after the second step-pulse application.

Panels I-K:

Provide additional traces. In Panel I, the inhibition by zinc is clear, but in Panel J, the reduction appears less distinct and could be due to rundown or an artifact. Additional controls should clarify this.

Figure 2K presents the most representative trace among five recorded cells. The apparent reduction is less distinct, likely due to an artifact caused by a bubble in the rapid perfusion system during solution exchange. However, at the end of zinc application ($t = 50$ s), the current amplitude was clearly reduced compared with that at $t = 0$ –10 s.

Figure 3

Panel D:

Include additional data showing the transition to pH 6 and washout with pH 7.5, similar to the experimental design in Panels A and B.

We included additional data showing raw trace of the application of pH 6.0 in Figure 3D, also included the transition to pH 6 and washout with pH 7.5 in Figure 3E.

Figure 3-Supplement 1:

Include zinc washout experiments. This approach is one of the best ways to evaluate the reversibility of zinc inhibition on the channel.

As mentioned above, in this recording we recorded step pulses up to +180 mV. The zinc washout effect could not be quantified because the patch was usually lost after the second step-pulse application.

Figure 6

Zinc Inhibition Specificity:

The authors should use quinidine, a known washable Slo3 inhibitor, to assess Slo3 activity before and after zinc injection.

This experiment would confirm that zinc specifically inhibits Slo3, rather than affecting other endogenous channels.

We sincerely thank the reviewer for this valuable suggestion. However, given the technical difficulty of these experiments, which involve lengthy VCF recordings and manual zinc injections that significantly compromise oocyte health, it is not feasible to apply quinidine at this stage.

Moreover, we observed voltage-dependent fluorescence changes around the VSD, and this change was influenced by the application of zinc, confirming that zinc specifically inhibits Slo3 rather than affecting other endogenous channels.

Discussion - Key Revisions Needed

Line 308: "Our results demonstrated that intracellular zinc is exported from spermatozoa during capacitation."

This claim needs to be supported by experiments using non-capacitated conditions.

Additionally, measuring maximum and minimum zinc concentrations under different conditions would improve the interpretation of fluorescence intensity changes.

We now include negative control in non-capacitated sperm. The data is incorporated into Figure 1B.

Line 309: "We further discovered that intracellular zinc regulates alkalization-induced hyperpolarization in mice spermatozoa, mediated by Slo3 channel."

Additional controls are needed to substantiate this claim.

At this stage of the study, we do not have access to Slo3 knockout (KO) mice; therefore, performing additional experiments is not feasible.

Line 316: "Using FluoZin3-AM for zinc imaging, we confirmed the presence of intracellular zinc in sperm (Fig. 1A), which is consistent with previous findings (Henkel et al., 1999). Our observations revealed that treatment with capacitation medium induced a decrease in zinc fluorescence intensity (Fig. 1B, C), suggesting that zinc levels are dynamic during capacitation."

This statement must be supported by negative controls, including non-capacitated sperm conditions.

We now include negative control in non-capacitated sperm. The data is incorporated into Figure 1B.

Line 327: "We also observed that zinc chelator significantly affected the sperm motility only after, but not before, capacitation (Fig. 1-figure supplement 1)."

Data presentation should be revised to highlight the effects of capacitation itself.

The discussion should specify which motility parameters were affected and why others were not.

In the text we mentioned that:

"We incubated the isolated spermatozoa with cell permeable Zn^{2+} chelator N,N,N',N'-Tetrakis(2-pyridylmethyl)ethylenediamine (TPEN) and measured the motility parameters before and after capacitation. We found that VAP (average path velocity), VCL (curvilinear velocity), and VSL (straight-line velocity) were influenced by the TPEN treatment only after the capacitation, as shown in Fig. 1—figure supplement 1. These results demonstrate that the dynamics of zinc levels during capacitation potentially contributes to sperm motility, highlighting the importance of zinc action in sperm physiology."

Indeed, we observed that zinc chelator significantly affected the sperm motility specifically in VAP (average path velocity), VCL (curvilinear velocity), and VSL (straight-line velocity) only after, but not before, capacitation (Fig. 1—figure supplement 1). Of note, it has been recently

reported that all these motility parameters (VAP, VCL, and VSL) are reduced by Slo3-specific inhibitors in human sperm (M. Lyon et al., 2023). These findings are consistent with the idea that endogenous zinc dynamics control sperm motility through Slo3 during the capacitation process.

Figure legend is revised accordingly.

Line 369: "Structural determinants of zinc inhibition in the mSlo3 channel."

The authors should include an analysis of the evolutionary conservation of the mutated sites across Slo1, Slo2, and Slo3.

If Slo3 has a unique regulatory mechanism, these sites should show high sequence variability compared to other Slo channels.

If these sites are highly conserved, the authors should explain how Slo3 differs functionally from Slo1 and Slo2 despite this conservation.

We thank the reviewer for the valuable suggestions regarding the inclusion of additional discussion points on the structural determinants of zinc inhibition in the mSlo3 channel. We performed sequence alignment by using ClustalO between mSlo3, mSlo1, and mSlo2.2. It is worth noting that only human and frog variants of Slo2.1 sequence are available in the database, so we included only Slo2.2 subtype, as our focus was on Slo3 in mouse sperm.

Based on the alignment, E169 (mSlo3 numbering) is conserved among the Slo family channels in mice, while in contrast E205 (mSlo3 numbering) is not. To date, there have been no report examining the corresponding residues to E169 (E191 in mslo1 or E176 in mslo2.2) for their zinc sensitivity. This might be because in both channels the zinc-binding sites are well defined where they are located in RCK1 domain for Slo1 (Hou et al., 2010) and RCK2 domain for Slo2.2 (J. Zhang et al., 2023). The identified binding site in Slo2.2 is conserved in Slo2.1 but not present in Slo1 and Slo3 (J. Zhang et al., 2023), further suggesting that zinc regulation differs among Slo family members. However, this does not rule out the possibility that regions surrounding E191 or E176 could provide to additional insights into zinc regulation in these channels, which could be of interest for future studies.

Interestingly, in contrast to E169, E205 is not conserved across the Slo family, making this residue unique to the mouse Slo3 channel and potentially a determinant of zinc sensitivity in mSlo3. Given that E205 is located in the S4 domain and supported by our VCF results showing that zinc inhibition influences the motion of voltage-sensing domain of mSlo3, E205 represents an important residue to be explored in future studies. Furthermore, as this residue is unique only to Slo3, it highlights the distinct functional properties of Slo3 such as its gating mechanism as it is regulated by both membrane voltage and alkalinization, which has a different voltage range of activation compared to mSlo1 (Li et al., 2024) and involves distinct ligands and gating mechanisms compared to Slo2 (J. Zhang et al., 2023).

We add the sequence alignment results into Figure 5—figure supplement 1F.

We revised the results section as follows:

“Additionally, we performed sequence alignment by using ClustalO between mSlo3, mSlo1, and mSlo2.2. It is worth noting that only human and frog variants of Slo2.1 sequence are available in the database, so we included only Slo2.2 subtype, as our focus was on Slo3 in mouse sperm. Based on the alignment, E169 (mSlo3 numbering) is conserved among the Slo family channels in mice, while in contrast E205 (mSlo3 numbering) is not. (Figure 5—figure supplement 1F).”

We revised the discussion section as follows:

“Based on sequence alignment, E169 (mSlo3 numbering) is conserved among Slo family channels in mice, whereas E205 (mSlo3 numbering) is not (Fig. 5—figure supplement 1F). To date, no studies have examined the corresponding residues to E169 (E191 in mSlo1 or E176 in mSlo2.2) for their potential zinc sensitivity, likely because the established zinc binding sites in these channels are located in the RCK1 domain for Slo1 (Hou et al., 2010) and the RCK2 domain for Slo2.2 (J. Zhang et al., 2023). The identified zinc binding site in Slo2.2 is conserved in Slo2.1 but is absent in both Slo1 and Slo3 (J. Zhang et al., 2023), further suggesting that zinc regulation differs among Slo family members. Although regions surrounding E191 or E176 may still provide additional insights into zinc regulation and could be of interest for future investigation, E205 stands out because, unlike E169, it is not conserved across the Slo family, making it unique to mSlo3 and potentially a specific determinant of zinc sensitivity in this channel.”

Figure legend is revised accordingly.

Line 392: "Physiological relevance of zinc inhibition of the mSlo3 channel in mouse sperm."

The authors should mention the effects of zinc on CatSper channels, as CatSper is also crucial for capacitation.

Slo3 inhibition may represent only one component of zinc's broader regulatory role during capacitation.

We thank the reviewer for raising this important point regarding the physiological relevance of zinc inhibition of the mSlo3 channel in mouse sperm. We agree that we should have also discussed the effect of zinc on CatSper channels, as this channel is crucial for capacitation. To date, there are only few reports on the effect of zinc on CatSper channels, and none have been reported in mouse sperm. One important study was reported by (Jeschke et al., 2021), in which seminal zinc was found to inhibit prostaglandin-induced activation of CatSper in human sperm. The complex opposing action of seminal zinc and prostaglandins on CatSper may help preventing premature activation of CatSper in the ejaculate and act as a dilution sensor, which facilitating sperm to escape into female genital tract (Jeschke et al., 2021). Taking this into consideration, as the reviewer pointed out, zinc inhibition on Slo3 may represent only one component of zinc's broader regulatory role during capacitation.

We added a sentence to the discussion section in the revised manuscript as follows:

“In addition to that, to date, there are only few reports on the effect of zinc on other sperm ion channels, and none have been reported in mouse sperm. One important study was reported by (Jeschke et al., 2021), in which seminal zinc was found to inhibit prostaglandin-induced activation of CatSper, a sperm-specific Ca^{2+} channel, in human sperm. The complex opposing action of seminal zinc and prostaglandins on CatSper may help preventing premature activation of CatSper in the ejaculate and act as a dilution sensor, although this study does not provide direct evidence for zinc acting directly on CatSper (Jeschke et al., 2021).”

The study presents valuable insights into the role of intracellular zinc in sperm capacitation and Slo3 channel function. However, the physiological impact of these findings remains unclear due to insufficient controls and missing key experimental data. The suggested revisions would strengthen the validity of the claims made by the authors and improve the overall scientific rigor of the manuscript.

Key Areas for Improvement:

Control experiments in non-capacitated conditions.

Increased statistical rigor in figure analyses.

More detailed experiments to confirm specificity of zinc action on Slo3.

Expanded discussion of zinc's role beyond Slo3, including CatSper regulation.

The authors should measure these effects in sperm cells using the patch-clamp technique to directly record Slo3 currents. By normalizing Slo3 currents to cell capacitance at different intracellular zinc concentrations, the authors can quantitatively assess the extent of Slo3 inhibition by zinc and strengthen the physiological relevance of their findings.

By addressing these concerns, the manuscript will provide a more robust foundation for understanding zinc's regulatory role in sperm physiology and capacitation.

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